



LCR LCP API Internal Messages used by:

SR200/SR202/SR204/SR205/SR206/SR209

SR210/SR211/SR212/SR214/SR215/SR216/SR217/SR219

SR260/SR262/SR264/SR265/SR266/SR267/SR269

SR600/SR601

SR1000

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1 Functional Specifications

The Liquid Controls Protocol (LCP) is a general purpose protocol that is to be used in products designed and developed by Liquid Controls, LLC. It provides the general means of communications between a variety of devices over an RS-232, RS-422, or RS-485 hardware link or through a TCP/IP connection.

1.1 General Requirements

The following list shows the requirements of LCP.

- Any device using LCP can be the master, a slave, or both.
- Message collision control is performed via timing. To date, there has been no need to implement this feature as all Liquid Controls LCP devices have only been controlled by one master device. The details of this specification, as well as its implementation, will be addressed when needed.
- For serial communication, supported baud rates will be 2400, 4800, 9600, 19200, 57600, or 115200. Note that the LCR-II boards do not support 115200 baud.
- Characters will be transmitted using one start bit, 8 data bits, and one stop bit. No parity bit is transmitted. It is assumed that the CRC following each message will ensure data integrity.

1.2 LCP Message Format

In the message format described below, each sequence of characters enclosed in angle brackets represents one 8-bit byte which can contain binary data in the range of 0 to 255 decimal.

LCP messages use the following basic format:

```
~~<to><from><status><len><data0>...<datalen-1><crc0><crc1>
```

<to> Destination node address to which the message is being sent. This number can range from 0 to 255. Node address 0 is used to broadcast messages to all nodes in the network in which case no response message should be expected by the node that initiated the message sequence.

<from> Source node address of the message that is being sent. This number can range from 1 to 255.

<status> Status of message. The status byte looks as follows:

Bit	Description	
0	Message Identifier. This bit will toggle between 0 and 1 so that the slave device can distinguish between new messages and repeat messages.	
1	Synchronization. If this bit is set, the message received will be processed as a new message regardless of the state of the message identifier bit. This bit should be set once in an LCP session with a node so as to synchronize with the target node. Note that it is <i>not</i> the intention that this bit be used for all messages as you will lose the ability to have the slave device discard resent messages; using this bit with each message transmitted to a node is done at your own risk. Note that when the "Synchronization" bit is used, the "Message Identifier" bit is also saved by the target node; the next transmitted message should be sent with the "Synchronization" bit cleared, and the "Message Identifier" bit toggled.	
	Command Messages (Bit 7 = 0)	Response Messages (Bit 7 = 1)
2	Check Request. Set if the host wants to check on the status of the previous request that was indicated delayed by the slave device via the busy bit.	Busy. Set by the slave device to indicate to the host that it can't respond to the request but will do so as soon as possible.
3	Abort Request. Set if the host wants to abort a delayed request.	Request Aborted. Set if the slave aborted an active delayed request.
4	Reserved. Should be set to 0.	No Request Active. Set if the slave receives a check request and there is no delayed request active.
5	Reserved. Should be set to 0.	Buffer Overrun. Set if the receive buffer in the slave device was not large enough to contain the entire message.
6	Reserved. Should be set to 0.	Not Supported. Set if the command in the message received by the slave is not supported by the slave device.
7	Message Type. Set to 0 indicating the message is a command message from a host.	Message Type. Set to 1 indicating the message is a response message from a slave.

<len> Length of the data portion of the message.

<data_i> Data byte number i. The contents of the data portion of the message is device specific; see Section 2 for the details of the messages used by the various LCR registers.

<crc₀> Low order byte of the CRC.



<crc₁> High order byte of the CRC.

If a ~ character is to be transmitted in any field other than the two leading characters, it will be transmitted as <esc>~. This requires that the <esc> character be transmitted as <esc><esc>. This rule ensures that two sequential ~ characters can only occur at the start of a message allowing the ~~ to be used as a message synchronizing mechanism. The CRC will include any additional <esc> characters in its calculation with the exception of additional <esc> characters in the CRC itself, however the receiving device should replace the <esc> sequences with the appropriate single character before processing the message. Additional <esc> characters contained in the data portion of a message will not be included in the length field.

The CRC of the message is calculated using the following algorithm:

1. Use a 16-bit variable for the CRC and initialize it to 0. Using each byte of the LCP message in the order received or transmitted, apply steps 2 through 5.
2. Using each bit of the current byte starting with the most significant bit, apply steps 3 through 5.
3. Shift the CRC left one bit and save the resulting carry flag.
4. Logically OR the current bit into the least significant bit of the CRC.
5. If the carry flag saved in step 3 is a one, logically Exclusive-OR the CRC with the hexadecimal value 1021h.

The maximum length of a transmitted LCP message is 523 bytes which, when converted to its original contents would contain 263 bytes. Stripping off the 8 bytes of LCP overhead, this leaves 255 bytes maximum for the data portion of the message.

The following code shows how to build an LCP message using the C programming language. It includes the insertion of <esc> sequences as well as CRC calculation.

```

/*=====*/
/* Source File: Support.c */
/* Program: Generic */
/* Function: LCPBuildMessage */
/* */
/* Authors      Dates      Revision  Change Log */
/* Bryan Haynes  09/21/00    1.00    Original */
/* */
/* Function: */
/* The following section of code contains definitions, function */
/* prototypes, and global variable declarations. Typically, this code */
/* would reside in header files that would be included into the source */
/* files that use the LCPBuildMessage function. */
/* */
/* Inputs: */
/* None */
/* */
/* Outputs: */
/* None */
/* */
/*=====*/

// General definitions taken from ASCIIctl.h and LCP.h.

#define ESC ((char)'\x1B') // ASCII escape
#define LCP_MsgSizeLimit (523U) // maximum size of an LCP message

// Function prototypes taken from Support.h.

unsigned long LCPBuildMessage(unsigned char to, unsigned char status,
                             unsigned char *msg, unsigned char len);

// Static function prototypes.

static void LCPAppendByte(unsigned char byte, unsigned char **LCPMsgPtrPtr,
                          unsigned short *msgLen, unsigned short *crc);
static void LCPUpdateCRC(unsigned short *crc, unsigned char byte);

// Global variables.

unsigned char LCPHostNodeAddress = 255; // host node address
unsigned char LCPMsg[LCP_MsgSizeLimit]; // raw LCP message buffer

```



```

/*=====*/
/* Source File: Support.c */
/* Program: Generic */
/* Function: LCPBuildMessage */
/*
/*
/* Authors      Dates      Revision  Change Log
/* Bryan Haynes 09/21/00    1.00    Original
/*
/*
/* Function:
/* This function builds a message that can be sent using LCP. The
/* resulting message is stored in the static variable LCPMsg.
/*
/*
/* Inputs:
/* to      - Node address to which the message is being sent.
/* status  - Status byte to include in the message.
/* msg     - Pointer to the data portion of the message.
/* len     - Number of bytes in data portion of the message.
/*
/*
/* Outputs:
/* LCPBuildMessage - Number of bytes in the resulting LCP message.
/*
/*
/*=====*/

// LCPBuildMessage entry point.

unsigned short LCPBuildMessage(unsigned char to, unsigned char status,
                               unsigned char *msg, unsigned char len)
{
    // Local variables.

    unsigned char *LCPTxMsgPtr;    // pointer to the LCP transmit message buffer
    unsigned short msgLen;         // absolute message length
    unsigned short crc;           // CRC of message
    unsigned char i;              // loop variable

    // Build the LCP message header.

    LCPTxMsgPtr = &LCPMsg[0];    // initialize pointer to the LCP transmit buffer

    *LCPTxMsgPtr++ = '~';         // insert the first ~
    *LCPTxMsgPtr++ = '~';         // insert the second ~
    msgLen = 2U;                  // length of LCP message
    crc = 0x7E7E;                 // seed the CRC with the message header

    // Build the variable part of the message.

    LCPAppendByte(to, &LCPTxMsgPtr, &msgLen, &crc);
    LCPAppendByte(LCPHostNodeAddress, &LCPTxMsgPtr, &msgLen, &crc);
    LCPAppendByte(status, &LCPTxMsgPtr, &msgLen, &crc);
    LCPAppendByte(len, &LCPTxMsgPtr, &msgLen, &crc);

    for (i=0; i < len; i++)      // loop to include the message
        LCPAppendByte(msg[i], &LCPTxMsgPtr, &msgLen, &crc);

    LCPAppendByte((unsigned char)(crc%0x0100), &LCPTxMsgPtr, &msgLen, NULL);
    LCPAppendByte((unsigned char)(crc/0x0100), &LCPTxMsgPtr, &msgLen, NULL);

    // LCPBuildMessage exit point.

    return(msgLen);              // return the absolute length of the message
}

```



```

/*=====*/
/* Source File: Support.c */
/* Program: Generic */
/* Function: LCPAppendByte */
/* */
/* Authors      Dates      Revision  Change Log */
/* Bryan Haynes 09/21/00    1.00     Original */
/* */
/* Function: */
/* This function appends a byte to the end of an LCP message. If the byte */
/* is an <esc> or ~ character, the two character escape sequence is */
/* appended, i.e. <esc><esc> and <esc>~ respectively. If the pointer to */
/* the CRC is not NULL, the CRC is updated accordingly. */
/* */
/* Inputs: */
/* byte - Byte that is to be appended to the LCP message. */
/* LCPMsgPtrPtr - Pointer to the LCP message pointer. */
/* msgLen - Pointer to the length of the LCP message. */
/* crc - Pointer to the CRC of the LCP message. */
/* */
/* Outputs: */
/* None */
/* */
/*=====*/

// LCPAppendByte entry point.

static void LCPAppendByte(unsigned char byte, unsigned char **LCPMsgPtrPtr,
                          unsigned short *msgLen, unsigned short *crc)
{
    // Check for special characters.

    if (byte == ESC || byte == '~') // check for special characters
    {
        *((*LCPMsgPtrPtr)++) = ESC; // insert the <esc> character
        (*msgLen)++; // update the LCP message length
        LCPUpdateCRC(crc, ESC); // include the <esc> in the CRC
    }

    // Append the byte to the end of the message.

    *((*LCPMsgPtrPtr)++) = byte; // insert the new byte
    (*msgLen)++; // update the LCP message length
    LCPUpdateCRC(crc, byte); // include the byte in the CRC

    // LCPAppendByte exit point.
}

```



```

/*=====*/
/* Source File: Support.c */
/* Program: Generic */
/* Function: LCPUpdateCRC */
/* */
/* Authors      Dates      Revision  Change Log */
/* Bryan Haynes 07/27/00    1.00     Original */
/* */
/* Function: */
/* This function updates the CRC of an LCP message with the value of the */
/* received byte. */
/* */
/* Inputs: */
/* crc - Pointer to the CRC of the LCP message. */
/* byte - Byte that is to be included in the CRC. */
/* */
/* Outputs: */
/* None */
/* */
/*=====*/

// LCPUpdateCRC entry point.

static void LCPUpdateCRC(unsigned short *crc, unsigned char byte)
{
    // Local variables.

    char i; // loop variable
    char XORFlag; // flag indicating the polynomial should be XORed

    // Combine the new data byte with the current CRC.

    if (crc != NULL) // ensure the CRC is wanted
    {
        for (i=7; i >= 0; --i) // loop through each bit in the data byte
        {
            XORFlag = (unsigned char)((*crc & 0x8000) != 0x0000);

            *crc <<= 1; // shift the CRC left
            *crc |= (unsigned short)((byte >> i) & 0x01);

            if (XORFlag) // check for CRC overflow
                *crc ^= 0x1021; // XOR with polynomial
        }
    }

    // LCPUpdateCRC exit point.
}

```

1.3 Example LCP Messages

When starting a communications session with an LCR, it is a good idea to ensure that the device being addressed is actually an LCR. This can be done by sending the LCP device the generic “Get Product ID” message which all LCP devices support. This is also a good time to synchronize with the LCR by setting the “Synchronization” bit in the status byte of the LCP message. The message to get the product ID of LCP node address 250 and force a synchronization with the host application that is using node address 255 would look as follows:

Transmit (LCP): <7Eh><7Eh><FAh><FFh><02h><01h><00h><2Fh><34h>

The response from the LCR will look similar to the following:

Receive (LCP): <7Eh><7Eh><FFh><FAh><80h><0Dh><00h><02h><53h><52h>
 <32h><30h><30h><62h><32h><2Eh><30h><35h><00h><1Fh>
 <77h>

This response message shows that the host is being addressed from LCR node 250, the status byte has bit 7 set indicating that the message is a response message, the length of the data portion of the message is 13 decimal, the return code is 0 indicating no errors occurred, the Product ID is 2 indicating the device is indeed an LCR, and the ASCII string containing the product name and revision is “SR200b2.05”. The CRC validates that the message was transmitted successfully. Note that the product name and revision string will almost certainly be different than that shown in this example.

The transmit message above addresses node 250 which is the default node address of LCR registers shipped from Liquid Controls. The two messages described above would look as follows if the LCR node address is 1 instead of 250.

Transmit (LCP): <7Eh><7Eh><01h><FFh><02h><01h><00h><8Ch><27h>

Receive (LCP): <7Eh><7Eh><FFh><01h><80h><0Dh><00h><02h><53h><52h>
<32h><30h><30h><62h><32h><2Eh><30h><35h><00h><2Eh>
<36h>

Once it has been determined that the LCP node is an LCR, other LCP02 messages can be sent to the device. For example, to get the version number of the LCP driver in LCR node address 1, the following message can be sent. Note that the synchronization bit has been cleared and the message identifier bit has been toggled in the status byte.

Transmit (LCP): <7Eh><7Eh><01h><FFh><01h><01h><26h><C9h><17h>

The response from the LCR will look as follows:

Receive (LCP): <7Eh><7Eh><FFh><01h><81h><04h><00h><21h><01h><00h>
<15h><24h>

This message shows that message ID 1 is being responded to, the length of the data portion of the message is 4 decimal, the return code is 0 indicating no errors occurred, the machine status byte is 21h which can be analyzed using the defines in LCP02.h, and the version number of the LCP driver in the LCR is 1.00. The CRC validates that the message was transmitted successfully. Note, again, that the version number of the LCP driver will almost certainly be different than that shown in this example.



1.4 LCR.iQ Considerations

The LCR.iQ is the next step in the evolution of the LCR family of electronic registers. While there are numerous new features that are available in the LCR.iQ, with all change comes the side effect that some features have been relegated to history's dustbin.

First, the good news. There are currently 31 new fields of data that the LCR.iQ provides to the third-party developer, and this list will certainly grow as new features are added to the register. These fields are accessible using the "Get Extended Field Data" message (0x40) and the "Set Extended Field Data" message (0x41) as shown in section 2 of this document. You can refer to the "Field Numbers" table in section 2 to see the complete list of fields supported by the various LCR registers. Note that the original LCR field set can be accessed using these two new functions whereas the new LCR.iQ fields cannot be accessed via the original "Get Field Data" (0x20) and "Set Field Data" (0x21) messages.

Following is a list of fields that have been provided in earlier LCR's that are not supported by the LCR.iQ. Note that in later releases of the LCR.iQ software, some of these fields may be supported and thus removed from the list of unsupported fields.

Unsupported Fields in the LCR.iQ			
Field	Name	Type	Description
9	PulsesPerDistance_UL	UFLOAT+4	Contains the number of pulses that have to occur for each distance unit displayed on the odometer. Currently not in use.
10	CalibDistance_UL	UFLOAT+1	Calibrated odometer reading. This field can be set to the actual distance traveled so as to calculate the correct value for Field #9. Currently not in use.
12	Odometer_UL	UFLOAT+1	Odometer reading. Currently not in use.
112	DBMNode	INTEGER	The node address of the Database Manager application running on the office PC (LCR600 only).
113	FSNode	INTEGER	The node address of the File Server running in the LCR (LCR600 only).
122	DeliveryScreen	LIST+53	Format of the delivery screen (LCR600 only).
123	BootloaderRev	TEXT	Bootloader revision number (LCR600 only).
124	ActivateFeatureKey	TEXT	The key used to activate the advanced feature. (LCR600 only)
125	FeaturesActivated	LONG	The set of flags that indicate which features have been purchased. (LCR600 only)
128	AvgFlowBase	LIST+54	Specifies the time unit that is applied when calculating the flow rate.
130	AvgDensity	UFLOAT+4	The Average density of the fuel during delivery.
131	DensityUnits	TEXT	Specifies the unit that is used for density measurement.
192	LCR 600 POS fields.		
.			
.			
.			
240			

The design change in the LCR.iQ that will have the largest impact on some third-party software is the removal of the **Red Switch**. The functionality of the red switch is now performed with the soft keys of the LCR.iQ; for example deliveries are started from the “Home” screen with the F5/Start key.

When a Run command is sent to an LCR, LCR-II, or LCR 600 and the switch is in the Run position, the delivery will be started immediately. If the switch is not in the Run position when the Run command is sent, the command will be queued in the register so that when the switch is moved to Run, the delivery will start. The primary use of the switch position information in the device status byte is to provide a way for the third-party developer to inform the fueler to move the switch to the Run position in order to start the next delivery in the event that the switch is in the wrong position. Because the red switch has been removed from the LCR.iQ, an attempt has been made to emulate the switch position so as to fill the needs of the majority of third-party software.

The most interesting cases, and the ones that should be validated by the third-party developer, are the ones where tickets are being printed, specifically the delivery ticket and the shift ticket. If the print commands come from the register by one of the soft keys, the switch position will show as “Print” or “Shift Print” during the printing of the ticket. Once the ticket has been printed, it will automatically revert back to the “Run” position. The details can be ascertained by studying the following algorithm.

LCR.iQ Switch Position Emulation Algorithm

Preamble: When power is applied to the register, the switch position will be set to “Run” and the source of the last command will be set to “Register.” The following algorithm will then be run as part of the power up initialization process.

The algorithm will also be run periodically at the end of the register state machine to ensure that the latest status bits are reflected properly in the switch position.

1. If the W&M bolt is removed, set the switch position to “Calibrate” and go to step 11.
2. If the current register state is not a paused delivery, set the switch position to “Run.”
3. If the source of the last register command is from an LCP host application, go to step 11.
4. If a delivery is paused, go to step 10.
5. If a delivery is active, go to step 11.
6. If a delivery ticket is printing, go to step 9.
7. If a shift ticket is not printing, go to step 11.
8. Set the switch position to “Shift Print” and go to step 11.
9. Set the switch position to “Print” and go to step 11.
10. Set the switch position to “Stop.”
11. Terminate the algorithm.

The position of the red switch is contained in the device status byte which is returned in most LCP messages to the LCR.



LCP02 Messages that Return the Device Status	
Message Name	Message ID
Get Field Data	20h
Set Field Data	21h
Print Text on LCR Printer	22h
Get Machine Status	23h
Issue Command	24h
Set Device Address	25h
Get Version Number	26h
Get Security Level	27h
Get Delivery Status	28h
Get Extended Field Data	40h
Set Extended Field Data	41h
Get LCR.iQ Field Parameters	7Ah
Get LCR 600 Field Parameters	7Bh
Set Baud	7Ch
Get LCR-II Field Parameters	7Fh

If the developer was using the switch position to determine the state of the register, there is a more robust way of performing this functionality by using the state sub-field in the device status byte and/or the status bits returned in the “Get Machine Status” and “Get Delivery Status” messages.

The bits in the Delivery Code word contain the most useful information regarding the status of a delivery. By periodically checking these bits, the host application can determine if a new delivery has been queued in the register, if a new delivery is starting, if a delivery is active, if flow is active, and if a delivery ticket is pending. Using these bits to determine when a delivery is running or when it ends is advantageous over using the switch position because deliveries can end for reasons other than the switch. For example, a no-flow condition or a preset being reached can end a delivery, along with various error conditions. **The third-party developer is encouraged to use the position of the red switch only for generating instructions to the fueler and to use the Delivery Code bits to made decisions about the delivery state of the register.**

Delivery Code Bits		
Value	Name	Description
0x0001	LCRSc_DEL_TICKET_PENDING	Indicates that a delivery ticket is pending. A new delivery cannot be started until this bit is cleared via successfully printing the last delivery ticket.
0x0004	LCRSc_FLOW_ACTIVE	Indicates flow is active during a delivery. The bit is turned on and off with the Delivery Code Bit 0x0008 but is also turned off when a delivery is paused and turned back on when it is resumed.
0x0008	LCRSc_DELIVERY_ACTIVE	Indicates that a delivery is active. The bit is turned on just before Delivery Code Bit #10 is turned off and is turned off at the end of the delivery.
0x0400	LCRSc_BEGIN_DELIVERY	Indicates that a delivery is in the process of being started. Once the delivery has been started successfully, this bit is turned off.
0x0800	LCRSc_NEW_DELIVERY_QUEUED	Indicates that a new delivery has been queued in the LCR. This condition occurs when a Command #0 is issued and the switch is in the STOP, PRINT, or SHIFT PRINT position.

Delivery Status Bits		
Value	Name	Description
0x0200	LCRSd_STOP_REQUEST	This bit is set if a delivery is active and Command #1 is issued. It is cleared when a delivery is started or resumed or when Command #2 or Command #6 has been issued.
0x0400	LCRSd_DELIVERY_END_REQUEST	Indicates a request has been issued to end the current delivery. This occurs when a delivery is active and Command #2 or Command #6 has been issued.

2 Data Portion of LCR Specific Messages

The field numbers, device status byte, printer status byte, delivery status word, delivery code word, command bytes, and security level are described in detail in the **LCP SDK** documentation or can be obtained from the defines in LCP02.h. The third-party developer is encouraged to use LCP02.h if possible during the application development process.

It is important to note that all integers, longs, and floats are stored in the LCR with the MSB at the lowest address and the LSB at the highest address. This is opposite of how the Intel 80x86 processor stores its numeric data. For example, a no-flow timer of 180 will be transmitted to the host as 00h followed by B4h. In an Intel machine, these bytes need to be swapped to B4h followed by 00h before being used by the application.

LCR volume fields are stored as long integers with an implied decimal point as defined by Field #39. Floating point numbers are stored using the IEEE-754 format.

Data Type Definitions		
Type	Name	Description
A	ASCII Character	Contains a one byte ASCII character. The number of characters stored in the field is Size .
AZ	ASCIIZ String	Contains a NUL terminated string of ASCII characters. The maximum length of the string is Size-1 .
D	Double Floating Point	Contains an eight byte floating point number in IEEE-754 format.
F	Floating Point	Contains a four byte floating point number in IEEE-754 format.
SB	Signed Byte	Contains a one byte integer in the range -128 to 127.
SI	Signed Integer	Contains an integer where the range is operating system dependent. On 16-bit machines, it is a two byte integer in the range -32768 to 32767. On 32-bit machines, it is a four byte integer in the range -2147483648 to 2147483647.
SL	Signed Long	Contains a four byte integer in the range -2147483648 to 2147483647.
SS	Signed Short	Contains a two byte integer in the range -32768 to 32767.
ST	Structure	Data element is a structure where the contents and size vary depending on the structure definition.
UB	Unsigned Byte	Contains a one byte integer in the range of 0 to 255.
UI	Unsigned Integer	Contains an integer where the range is operating system dependent. On 16-bit machines, it is a two byte integer in the range 0 to 65535. On 32-bit machines, it is a four byte unsigned integer in the range 0 to 4294967295.
UL	Unsigned Long	Contains a four byte integer in the range 0 to 4294967295.
UN	Union	Data element is a union where the contents and size vary depending on the union definition.
US	Unsigned Short	Contains a two byte integer in the range 0 to 65535.
V	LCR Volume	Contains a signed four byte integer in the range -2147483648 to 2147483647 with an implied decimal point defined by the decimals field in the record.

Get Product ID (Generic LCP Message)				
Offset	Size	Type	Name	Description
00	01	UB	msgID	Message ID to get the product ID. Set to 00h.
Total	01			

Return from Get Product ID				
Offset	Size	Type	Name	Description
00	01	UB	rc	Return code.
01	01	UB	productID	Product identification byte. The LCR will return 02h.
02	n	AZ	name	ASCIIZ string containing the product name and revision string. The string length of name should be no larger than 15 implying that n should be no larger than 16.
Total	n+02			

Get Field Data				
Offset	Size	Type	Name	Description
00	01	UB	msgID	Message ID to get field data. Set to 20h.
01	01	UB	fieldNum	Field number to get.
Total	02			

Return from Get Field Data				
Offset	Size	Type	Name	Description
00	01	UB	rc	Return code.
01	01	UB	devStatus	Device status byte.
02	n	?	fieldData	Field data. Size and type dependent on field number.
Total	n+02			

Set Field Data				
Offset	Size	Type	Name	Description
00	01	UB	msgID	Message ID to set field data. Set to 21h.
01	01	UB	fieldNum	Field number to set.
02	n	?	fieldData	Field data. Size and type dependent on field number.
Total	n+02			

Return from Set Field Data				
Offset	Size	Type	Name	Description
00	01	UB	rc	Return code.
01	01	UB	devStatus	Device status byte.
Total	02			

Print Text on LCR Printer				
Offset	Size	Type	Name	Description
00	01	UB	msgID	Message ID to print text. Set to 22h.
01	n	UB	prnText	Text to print.
Total	n+01			

Return from Print Text on LCR Printer				
Offset	Size	Type	Name	Description
00	01	UB	rc	Return code.
01	01	UB	devStatus	Device status byte.
Total	02			

Note that the following “Get Machine Status” message is identical to the “Get Delivery Status” message defined below with the exception that the printer status is returned. If the printer is off-line, a two second delay will occur during the “Get Machine Status” request whereas no such delay will occur during a “Get Delivery Status” request. If the printer status is not needed, use the “Get Delivery Status” message below.

Get Machine Status				
Offset	Size	Type	Name	Description
00	01	UB	msgID	Message ID to get machine status. Set to 23h.
Total	01			

Return from Get Machine Status				
Offset	Size	Type	Name	Description
00	01	UB	rc	Return code.
01	01	UB	devStatus	Device status byte.
02	01	UB	prnStatus	Printer status byte.
03	02	US	delStatus	Delivery status word.
05	02	US	delCode	Delivery code word.
Total	07			

Issue Command				
Offset	Size	Type	Name	Description
00	01	UB	msgID	Message ID to issue command. Set to 24h.
01	01	UB	command	Command byte.
Total	02			

Return from Issue Command				
Offset	Size	Type	Name	Description
00	01	UB	rc	Return code.
01	01	UB	devStatus	Device status byte.
Total	02			



Set Device Address				
Offset	Size	Type	Name	Description
00	01	UB	msgID	Message ID to set device address. Set to 25h.
01	01	UB	newAddr	New device address.
Total	02			

Return from Set Device Address				
Offset	Size	Type	Name	Description
00	01	UB	rc	Return code.
01	01	UB	devStatus	Device status byte.
Total	02			

Get Version Number				
Offset	Size	Type	Name	Description
00	01	UB	msgID	Message ID to get version number. Set to 26h.
Total	01			

Return from Get Version Number				
Offset	Size	Type	Name	Description
00	01	UB	rc	Return code.
01	01	UB	devStatus	Device status byte.
02	02	UB	version	Major and minor version number of the LCR software.
Total	04			

Get Security Level				
Offset	Size	Type	Name	Description
00	01	UB	msgID	Message ID to get security level. Set to 27h.
Total	01			

Return from Get Security Level				
Offset	Size	Type	Name	Description
00	01	UB	rc	Return code.
01	01	UB	devStatus	Device status byte.
02	01	SB	security	Security level in the LCR.
Total	03			

Note that the following “Get Delivery Status” message is identical to the “Get Machine Status” message defined above with the exception that the printer status is not returned. If the printer is off-line, a two second delay will occur during the “Get Machine Status” request whereas no such delay will occur during a “Get Delivery Status” request. If the printer status is not needed, this is the message to use.

Get Delivery Status				
Offset	Size	Type	Name	Description
00	01	UB	msgID	Message ID to get machine status. Set to 28h.
Total	01			

Return from Get Delivery Status				
Offset	Size	Type	Name	Description
00	01	UB	rc	Return code.
01	01	UB	devStatus	Device status byte.
02	02	US	delStatus	Delivery status word.
04	02	US	delCode	Delivery code word.
Total	06			

Activate Pump & Print				
Offset	Size	Type	Name	Description
00	01	UB	msgID	Message ID to activate Pump & Print mode. Set to 29h.
Total	01			

Return from Activate Pump & Print				
Offset	Size	Type	Name	Description
00	01	UB	rc	Return code.
Total	01			

Get Transaction Record (SR266)				
Offset	Size	Type	Name	Description
00	01	UB	msgID	Message ID to get the oldest transaction record. Set to 2Ah.
Total	01			

Return from Get Transaction Record				
Offset	Size	Type	Name	Description
00	01	UB	rc	Return code.
01	04	F	temperature	Temperature of product at the end of the delivery.
05	04	UL	customerID	Customer ID number.
09	04	SL	saleNumber	Sale number.
0D	04	V	gross	Gross volume delivered.
11	04	V	net	Net volume delivered.
15	02	US	status	Delivery status at the end of the delivery.
17	01	UB	compType	Active compensation type.
18	01	UB	dateFormat	Date format used in the date and time field.
19	01	UB	decimals	Decimal setting for the volume fields.
1A	01	UB	product	Product ID delivered.
1B	01	UB	qtyUnits	Unit of measure for the volume fields.
1C	01	UB	tempScale	Temperature scale.
1D	12	AZ	dateTime	Date and time at the end of the delivery.
Total	2F			

Delete Transaction Record (SR266)				
Offset	Size	Type	Name	Description
00	01	UB	msgID	Message ID to delete the oldest transaction record. Set to 2Bh.
Total	01			

Return from Delete Transaction Record				
Offset	Size	Type	Name	Description
00	01	UB	rc	Return code.
Total	01			

Get First Ticket Line				
Offset	Size	Type	Name	Description
00	01	UB	msgID	Message ID to get the first ticket line of the last delivery ticket. Set to 2Ch.
Total	01			

Return from Get First Ticket Line				
Offset	Size	Type	Name	Description
00	01	UB	rc	Return code.
01	n	AZ	ticketText	First ticket line text.
Total	n+01			

Get Next Ticket Line				
Offset	Size	Type	Name	Description
00	01	UB	msgID	Message ID to get the first ticket line of the last delivery ticket. Set to 2Dh.
Total	01			

Return from Get Next Ticket Line				
Offset	Size	Type	Name	Description
00	01	UB	rc	Return code.
01	n	AZ	ticketText	Next ticket line text.
Total	n+01			

Get Extended Field Data				
Offset	Size	Type	Name	Description
00	01	UB	msgID	Message ID to get field data. Set to 40h.
01	02	US	fieldNum	Field number to get.
Total	03			

Return from Get Extended Field Data				
Offset	Size	Type	Name	Description
00	01	UB	rc	Return code.
01	01	UB	devStatus	Device status byte.
02	n	?	fieldData	Field data. Size and type dependent on field number.
Total	n+02			

Set Extended Field Data				
Offset	Size	Type	Name	Description
00	01	UB	msgID	Message ID to set field data. Set to 41h.
01	02	US	fieldNum	Field number to set.
03	n	?	fieldData	Field data. Size and type dependent on field number.
Total	n+03			

Return from Set Extended Field Data				
Offset	Size	Type	Name	Description
00	01	UB	rc	Return code.
01	01	UB	devStatus	Device status byte.
Total	02			

All register field parameters can be retrieved with message ID 7Ah, 7Bh, or 7Fh. They are the field type, field width, and the edit security level of each field. These parameters can be retrieved by setting the parameter number to 0, 1, or 2 respectively. The block number should be set to 0 initially and incremented on repeated calls until the sum of the number of parameter data bytes received back in the return message equals the number of fields in the LCR, e.g.

LCR_NUM_FIELDS. Note that the LCR determines the size of the data blocks and can theoretically change between different versions of the LCR control software. Code written to utilize this message should be written in a generic fashion so as to work regardless of the block size.

Get LCR.iQ Field Parameters				
Offset	Size	Type	Name	Description
00	01	UB	msgID	Message ID to get field parameters. Set to 7Ah .
01	01	UB	param	Parameter number.
02	01	UB	block	Block number.
Total	03			

Return from Get LCR.iQ Field Parameters				
Offset	Size	Type	Name	Description
00	01	UB	rc	Return code.
01	01	UB	devStatus	Device status byte.
02	n	UB	data	Parameter data bytes.
Total	n+02			

Get LCR 600 Field Parameters				
Offset	Size	Type	Name	Description
00	01	UB	msgID	Message ID to get field parameters. Set to 7Bh .
01	01	UB	param	Parameter number.
02	01	UB	block	Block number.
Total	03			

Return from Get LCR 600 Field Parameters				
Offset	Size	Type	Name	Description
00	01	UB	rc	Return code.
01	01	UB	devStatus	Device status byte.
02	n	UB	data	Parameter data bytes.
Total	n+02			

Set Baud				
Offset	Size	Type	Name	Description
00	01	UB	msgID	Message ID to set baud. Set to 7Ch.
01	01	UB	baudIX	Index into baud rate table. 0 = 57600, 1 = 19200, 2 = 9600, 3 = 4800, and 4 = 2400.
Total	02			

Return from Set Baud				
Offset	Size	Type	Name	Description
00	01	UB	rc	Return code.
01	01	UB	devStatus	Device status byte.
Total	02			

The “Check Request” and “Abort Request” messages were the original design in the LCR register to handle the case of queued requests. The status bits described in the LCP message definition above that perform these same functions were added at a later date when it became clear that this was a common concern for any type of LCP device. To support backward compatibility issues in the family of LCR registers, the following two messages have continued to be supported, whereas the status bits in the LCP status byte may not be supported due to the age of the device being controlled by the host application. You are free to use either mechanism when processing queued requests, but if backward compatibility issues are a concern to your application, it is recommended that the “Check Request” and “Abort Request” messages be used.

Check Request				
Offset	Size	Type	Name	Description
00	01	UB	msgID	Message ID to check previous request. Set to 7Dh.
Total	01			

Return from Check Request				
Offset	Size	Type	Name	Description
00	01	UB	rc	Return code.
01	n	?	?	The message response returned will be for the queued request command if it was completed in the meantime.
Total	n+01			

Abort Request				
Offset	Size	Type	Name	Description
00	01	UB	msgID	Message ID to abort previous request. Set to 7Eh.
Total	01			

Return from Abort Request				
Offset	Size	Type	Name	Description
00	01	UB	rc	Return code.
Total	01			



All register field parameters can be retrieved with message ID 7Ah, 7Bh, or 7Fh. They are the field type, field width, and the edit security level of each field. These parameters can be retrieved by setting the parameter number to 0, 1, or 2 respectively. The block number should be set to 0 initially and incremented on repeated calls until the sum of the number of parameter data bytes received back in the return message equals the number of fields in the LCR, e.g.

LCR_NUM_FIELDS. Note that the LCR determines the size of the data blocks and can theoretically change between different versions of the LCR control software. Code written to utilize this message should be written in a generic fashion so as to work regardless of the block size.

Get LCR-II Field Parameters				
Offset	Size	Type	Name	Description
00	01	UB	msgID	Message ID to get field parameters. Set to 7Fh.
01	01	UB	param	Parameter number.
02	01	UB	block	Block number.
Total	03			

Return from Get LCR-II Field Parameters				
Offset	Size	Type	Name	Description
00	01	UB	rc	Return code.
01	01	UB	devStatus	Device status byte.
02	n	UB	data	Parameter data bytes.
Total	n+02			

Field Numbers			
Field	Name	Type	Description
0	ProductNumber_DL	LIST+0	Product number that is active.
1	ProductCode_DL	TEXT	Product code that can be displayed on the delivery ticket. Each product number can be assigned a unique product code. The product code set during configuration will be the default for that product after power up.
2	GrossQty_NE	VOLUME	Gross quantity of current delivery. This value can go negative, unlike Field #44.
3	NetQty_NE	VOLUME	Net quantity of current delivery. This field will only contain a non-zero value if the product is being temperature compensated. This value can go negative, unlike Field #45.
4	FlowRate_NE	VOLUME	Flow rate of the current delivery. The label on this value is Field #38 per Field #41.
5	GrossPreset_PL	VOLUME	Gross preset value set for the current delivery. Each product number can be assigned a unique gross preset value.
6	NetPreset_PL	VOLUME	Net preset value set for the current delivery. Each product number can be assigned a unique net preset value.
7	Temp_NE	SFLOAT+1	Current temperature of product being delivered. The label on this value is Field #35 and is calculated as follows: (Field #77 * Field #78) + Field #79 + Field #34.
8	Residual_WM	LIST+20	Indicates how residuals are handled in the quantity fields. Residuals can be rounded or truncated.
9	PulsesPerDistance_UL	UFLOAT+4	Contains the number of pulses that have to occur for each distance unit displayed on the odometer. Currently not in use.
10	CalibDistance_UL	UFLOAT+1	Calibrated odometer reading. This field can be set to the actual distance traveled so as to calculate the correct value for Field #9. Currently not in use.
11	ProductDescriptor_DL	TEXT	Product descriptor that can be displayed on the delivery ticket. Each product number can be assigned a unique product descriptor. The product descriptor set during configuration will be the default for that product after power up.
12	Odometer_UL	UFLOAT+1	Odometer reading. Currently not in use.
13	ShiftGross_NE	VOLUME	Gross quantity of current shift. Each product number maintains its own gross shift quantity. This value is reset to zero when a shift ticket is printed or when Field #16 is set to 0.
14	ShiftNet_NE	VOLUME	Net quantity of current shift. Each product number maintains its own net shift quantity. This value is reset to zero when a shift ticket is printed or when Field #16 is set to 0.

15	ShiftDeliveries_NE	LONG	Number of deliveries made during the current shift. Each product number maintains its own number of deliveries. This value is reset to zero when a shift ticket is printed or when Field #16 is set to 0.
16	ClearShift_DL	LIST+5	Setting this field to 0 causes the shift data to be cleared. If Field #37 is set to 0, the shift ticket must be successfully printed before the shift data is cleared. If Field #37 is set to 1, the shift ticket will be printed if the printer is on-line and ready to print, but in any event, the shift data is then cleared.
17	GrossTotal_WM	VOLUME	Cumulative gross quantity delivered of the current product.
18	NetTotal_WM	VOLUME	Cumulative net quantity delivered of the current product.
19	DateFormat_UL	LIST+3	Format to use when displaying the date.
20	Date_UL	DATE	Current date.
21	Time_UL	TIME	Current time.
22	SaleNumber_WM	LONG	Sale number assigned to the current delivery. The sale number is incremented at the start of a delivery.
23	TicketNumber_WM	LONG	Ticket number to be printed on the next delivery ticket. Setting this field to zero causes the ticket number to not be printed on the delivery ticket. The ticket number is incremented after it is printed on a delivery ticket or duplicate delivery ticket.
24	UnitID_UL	TEXT	LCR unit identifier that can be displayed on the delivery ticket.
25	NoFlowTimer_DL	INTEGER	Time, in seconds, that is allowed when pausing a delivery before the delivery is automatically ended and a delivery ticket printed. Setting this field to zero disables this feature and causes a message, indicating that there were multiple deliveries at one site, to be printed on the delivery ticket.
26	S1Close_WM	VOLUME	Used when delivering a preset amount to indicate when to close solenoid 1, thus causing the LCR to deliver at a reduced dwell flow rate. When delivering a preset amount and Field #92 or Field #93 is less than or equal to this field, solenoid 1 will close.

27	PresetType_DL	LIST+7	Type of preset being delivered when Field #5 or Field #6 is not zero. Clear presets are those that will end the delivery when the preset amount is reached and then clear the preset amount. Multiple presets will pause the delivery when the preset amount is reached and allow the same preset amount to be delivered again. In this case, the preset amount will be cleared when the delivery is explicitly ended with Command #2, or when it ends when the no flow timer times out. Retain presets are similar to clear presets with the exception that the preset amount is not cleared at the end of the delivery. Required presets will require a preset amount to be entered prior to any and all deliveries.
28	PulseOutputEdge_UL	LIST+9	Indicates whether a rising or falling edge is generated on the scaled pulse output when the quantity is incremented.
29	Header_AE	LIST+8	Indicates which header line is accessible via Field #30.
30	TicketHeaderLine_UL	TEXT	Header line text that will be printed on the delivery ticket. If the header line is blank, it will not be printed on the ticket. Header line 11 will only be printed if it is not blank and if Field #86 is set to 0. Header line 12 will only be printed if it is not blank and if Field #87 is set to 1.
31	PrintGrossAndParam_WM	LIST+5	Indicates whether or not the gross volume and compensation parameter is to be printed on the delivery ticket. Note that if Field #31 is set to 0, the settings of Field #285 and Field #286 are irrelevant; if Field #31 is set to 1, the two ticket text lines are controlled independently with Field #285 and Field #286.
32	VolCorrectedMsg_WM	LIST+5	Indicates whether or not the volume corrected message is to be printed on the delivery ticket.
33	Temp_WM	SFLOAT+2	Current temperature of product being delivered. The label on this value is Field #35 and is calculated as follows: (Field #77 * Field #78) + Field #79 + Field #34.
34	TempOffset_WM	SFLOAT+2	Value that is added to Field #33 and Field #7 so as to fine tune the accuracy of the temperature reading. This field allows the temperature to be fine tuned without entering factory calibration.
35	TempScale_WM	LIST+10	Unit of the temperature fields, Field #33 and Field #7.
36	MeterID_WM	TEXT	LCR meter identifier that can be displayed on the delivery ticket.
37	TicketRequired_WM	LIST+5	Used to indicate whether or not a delivery ticket is required. If this field is set to 0, a complete delivery ticket must be printed before the next delivery can be started. If the field is set to 1, a delivery ticket will be printed if there is a printer available online. And if the field is set to 2, a delivery ticket will never be printed.

38	QtyUnits_WM	LIST+4	The unit label applied to the quantity delivered.
39	Decimals_WM	LIST+14	Number of places past the decimal that are to be displayed in fields of type VOLUME.
40	FlowDirection_WM	LIST+21	Specifies the LCR's interpretation of forward flow through the meter.
41	TimeUnit_WM	LIST+11	Specifies the time unit that is applied when calculating the flow rate.
42	CalibrationEvent_NE	LONG	Number of calibration events that have occurred in the LCR. Calibration events include the following: Pulses Per Unit (k-Factor), Linearization Flow Rate, Linearization Percent Error and Activating Linearization.
43	ConfigurationEvent_NE	LONG	Number of configuration events that have occurred in the LCR. For Weights and Measures use only. Configuration events include the following: Sale Number, Gross Totalizer, Net Totalizer, Product Type, Compensation Type, Compensation Parameter, Base Temperature, Auxiliary Multiplier, Auxiliary Unit of Measure, S1 Close, Temperature Offset, Temperature Scale, Quantity Unit of Measure, Decimal Setting, Flow Rate Unit of Measure, Printer Type, Ticket Required Flag, Flow Direction, Meter ID, Ticket Number, Residual Processing, Print Gross and Parameter Flag, Print Volume Corrected Message Flag.
44	GrossCount_NE	VOLUME	Gross quantity of current delivery. Unlike Field #2, this value will not go negative. Its value will match the number displayed on the electronic counter on the LCR when temperature compensation is not active.
45	NetCount_NE	VOLUME	Net quantity of current delivery. Unlike Field #3, this value will not go negative. Its value will match the number displayed on the electronic counter on the LCR when temperature compensation is active and the LCR is not in calibration mode.
46	ProverQty_WM	SFLOAT+3	Gross quantity of current prover delivery. This field maintains a higher accuracy than Field #2 and can be set to the actual amount delivered so as to calculate the correct value for Field #47.
47	PulsesPerUnit_WM	UFLOAT+4	Contains the number of pulses that have to occur for each unit of product delivered.
48	AuxMult_WM	UFLOAT+3	Multiplier applied to the gross or net quantity delivered which results in an auxiliary quantity that can be printed on the delivery ticket or displayed to the user. The gross quantity is used if temperature compensation is not active, otherwise the net quantity is used.
49	AuxUnit_WM	LIST+6	The unit label applied to the auxiliary quantity delivered.
50	CalibrationNumber_NE	LONG	Number of times calibration mode has been entered in the LCR.

51	LinearPoint_AE	LIST+12	Linearization point number that is accessible via the fields Field #52 and Field #53.
52	LinearFlowRate_WM	UFLOAT+2	Flow rate of the current linearization point. A flow rate of 0.00 disables the use of the linearization point.
53	LinearError_WM	SFLOAT+3	Percent error of the quantity delivered when the product flows at the rate defined in Field #52.
54	LinearProverQty_WM	SFLOAT+3	Gross quantity of current prover delivery. This field can be set to the actual amount delivered so as to calculate the percent error stored in Field #53.
55	Linearize_WM	LIST+25	Used to specify that the linearization points should be applied to deliveries. When setting this field to 0 the linearization points are sorted by flow rate in descending order.
56	Printer_WM	LIST+19	Type of printer that is connected to the RS-232 printer port of the LCR.
57	CompensationType_WM	LIST+15	Compensation type that is being used for the current active product. Each product number can be assigned a unique compensation type.
58	CompensationParam_WM	FFLOAT	Compensation parameter that is being used for the current active product. Each product number can be assigned a unique compensation parameter.
59	BaseTemp_WM	SFLOAT+1	Base temperature used when calculating the net quantity based on the selected compensation type and parameter. Each product number uses a unique base temperature for that product. The base temperature can be set for linear compensation only.
60	Software_NE	TEXT	Version of the software running in the LCR.
61	PricePerUnit_DL	SFLOAT+4	Price charged per unit of delivered product.
62	TaxPerUnit_DL	SFLOAT+4	Tax charged per unit of delivered product.
63	PercentTax_DL	SFLOAT+4	Percent tax charged per monetary unit of delivered product.
64	DiagnosticMessages_AE	LIST+16	Used to get the next index for any active diagnostic messages. Calling the get function with zero in the field data causes the first diagnostic message index to be returned. If the get function is called without a zero in the field data, the first diagnostic message is returned if the delivery status or delivery code has changed since the last call. Calling the set function will always return the next diagnostic message index as a circular list. Note that the better way to get to these status bits is with the "Get Machine Status" message (MsgID 23h).
65	TotalTaxPerUnit_NE	SFLOAT+2	Total tax per unit charged for the delivered product. This is the rounded result of multiplying Field #2 or Field #3 by Field #62. Field #2 is used if temperature compensation is not active, otherwise Field #3 is used.
66	TotalPercentTax_NE	SFLOAT+2	Total percentage tax charged for the delivered product. This is the rounded result of multiplying Field #95 by Field #63.

67	ADCCode_NE	INTEGER	Contains a bit map showing how the temperature and voltage calibrations were performed. The LCR A to D Code Masks are used to mask the flags in this field.
68	SupplyVoltage_NE	UFLOAT+1	Supply voltage being applied to the LCR. The supply voltage is calculated as follows: (Raw Voltage * 0.00125) + 12.
69	PulserReversals_NE	LONG	Number of pulser reversals that have occurred during the current delivery.
70	ShiftStart_NE	TEXT	Date and time the current shift started.
71	AuxQty_NE	VOLUME	Auxiliary quantity of current delivery. This value is calculated by multiplying Field #2 or Field #3 by Field #48. Field #2 is used if temperature compensation is not active, otherwise Field #3 is used.
72	UserKey_DL	TEXT	Used to unlock the LCR or change the users key. If the LCR is currently locked and this field receives the key that matches the one in Field #81, the LCR will be unlocked. If the LCR is already unlocked, any value set in this field becomes the new user key by being set in Field #81.
73	Security_UL	LIST+13	Getting this field will return the current locked/unlocked status of the LCR. Setting this field to 0 will lock the LCR.
74	FactoryKey_AE	TEXT	Setting this field to the correct factory key places the LCR at factory security level. Factory security level is terminated when an incorrect factory key is set, when the switch on the LCR leaves the calibration position, or when power is applied to the LCR.
75	R100_0_FL	SFLOAT+1	Setting this field to any value causes it to be reset to the average of Field #77 read 100 times. This new value is then used to calculate Field #79 as follows: -Field #75 * Field #78. A $100.0 \pm 0.01\%$ ohm resistor must be attached to the temperature probe input before this field is set. Setting this field causes bit #2 and bit #3 to be cleared and bit #0 to be set in Field #67.
76	R128_6_FL	SFLOAT+1	Setting this field to any value causes it to be reset to the average of Field #77 read 100 times. This new value is then used to calculate Field #78 as follows: $74.0 / (\text{Field \#76} - \text{Field \#75})$. Field #79 is then recalculated as follows: -Field #75 * Field #78. Field #75 must be set before this field is set. A $128.6 \pm 0.01\%$ ohm resistor should be attached to the temperature probe input before this field is set. Setting this field causes bit #1 to be set in Field #67.
77	RawADC_NE	INTEGER	Raw ADC value coming from the A to D converter used to calculate the temperature of the product being delivered.

78	RTDSlope_FL	UFLOAT+4	Slope used when calculating the temperature of the product from the raw ADC. This field can either be set explicitly or be calculated by setting Field #75 followed by setting Field #76. Setting this field explicitly causes bit #0 and bit #1 to be cleared and bit #2 to be set in Field #67.
79	RTDOffset_FL	SFLOAT+3	Offset used when calculating the temperature of the product from the raw ADC. This field can either be set explicitly or be calculated by setting Field #75 followed by setting Field #76. Setting this field explicitly causes bit #0 and bit #1 to be cleared and bit #3 bit to be set in Field #67.
80	SerialID_FL	TEXT	Identifier of the hardware board in the LCR.
81	UserSetKey_FL	TEXT	Contains the users key that can be used to unlock the LCR.
82	LCRReset_FL	LIST+24	When set to 0, all fields in the LCR will be set to their initial default values.
83	CompParamType_NE	LIST+17	Compensation parameter that will be printed on the delivery ticket for the current active product. This is a parallel list to the one accessed via Field #57.
84	CompTempType_NE	LIST+18	Compensation temperature that will be printed on the delivery ticket for the current active product. This is a parallel list to the one accessed via Field #57.
85	PresetsAllowed_DL	LIST+1	Specifies the types of presets that can be used.
86	Aux1_DL	LIST+22	Defines how auxiliary port 1 should be used. Setting this to 0 causes header line 11 to be printed on the delivery ticket, assuming header line 11 is not blank.
87	Aux2_DL	LIST+23	Defines how auxiliary port 2 should be used. Setting this to 1 causes header line 12 to be printed on the delivery ticket, assuming header line 12 is not blank.
88	DefaultProduct_NE	LIST+0	Default product number used when the LCR is powered up.
89	DeliveryStart_NE	TEXT	Date and time the last delivery started.
90	DeliveryFinish_NE	TEXT	Date and time the last delivery finished.
91	LastCalibrated_NE	TEXT	Date and time of last calibration.
92	GrossRemaining_NE	VOLUME	Gross quantity remaining on the current gross preset delivery.
93	NetRemaining_NE	VOLUME	Net quantity remaining on the current net preset delivery.
94	ProductType_WM	LIST+2	Type of product being delivered. Each product number can be assigned a unique product type.
95	SubTotalCost_NE	SFLOAT+2	Non-taxed cost of delivered product. This is the rounded result of multiplying Field #2 or Field #3 by Field #61. Field #2 is used if temperature compensation is not active, otherwise Field #3 is used.
96	TotalTax_NE	SFLOAT+2	Total tax charged for the delivered product. This is the sum of Field #65 and Field #66.

97	TotalCost_NE	SFLOAT+2	Taxed cost of delivered product. This is the sum of Field #95 and Field #96.
98	Ticket_NE	TEXT	Version of the delivery ticket format being used in the LCR.
99	Language_NE	TEXT	Version of the language text being displayed on the lap pad in RS-232 mode.
100	PreviousGross	VOLUME	Gross meter totalizer at the beginning of the last delivery.
101	PreviousNet	VOLUME	Net meter totalizer at the beginning of the last delivery.
102	LCRNode	INTEGER	Node address used by the LCR when it is run in the 485 bank of software.
103		TEXT	Reserved—Should not be accessed by a host application.
104		TEXT	Reserved—Should not be accessed by a host application.
105	ShiftPassword	LONG	Password used to begin a new shift and/or delivery in SR266 control software.
106	CustomerID	LONG	Customer number receiving the fuel in SR266 control software.
107	UsePassword	LIST+26	Specifies when the password field will be required in SR266 control software.
108	UseCustomer	LIST+27	Specifies when a prompt will be issued for the customer number field in SR266 control software.
109	MaxPressure	SFLOAT+1	Maximum pressure differential read during the delivery in SR264 control software.
110	MaxPressureFlowRate	VOLUME	Flow rate where the maximum pressure differential was read during the delivery in SR264 control software.
111	ShutdownPressure	SFLOAT+1	The maximum allowed pressure reading from the transducer before the last delivery was terminated.
112	DBMNode	INTEGER	The node address of the Database Manager application running on the office PC (LCR600 only).
113	FSNode	INTEGER	The node address of the File Server running in the LCR (LCR600 only).
114	FSPort	LIST+50	Port address of the File Server (LCR600 only). Note in the LCR.iQ the base list for this field is LIST1000.
115	FSBaud	LIST+51	Baud rate used by the File Server (LCR600 only). Note in the LCR.iQ the base list for this field is LIST1000.
116	FSRetries	INTEGER	Number of retries used by the File Server (LCR600 only).
117	FSTxEnable	LIST+52	TxEnable bit used by the File Server (LCR600 only). Note in the LCR.iQ the base list for this field is LIST1000.
118	FSTimeout	INTEGER	Timeout used by the File Server (LCR600 only).
119	PulseOutputMultiplier	UFLOAT+2	Pulse Output Frequency for the current active product calibration (LCR600 only).
120	Aux1FlowRateToggle	VOLUME	Auxiliary 1 Flow Rate Toggle for the current active product calibration.

121	Aux2FlowRateToggle	VOLUME	Auxiliary 2 Flow Rate Toggle for the current active product calibration.
122	DeliveryScreen	LIST+53	Format of the delivery screen (LCR600 only).
123	BootloaderRev	TEXT	Bootloader revision number (LCR600 only).
124	ActivateFeatureKey	TEXT	The key used to activate the advanced feature. (LCR600 only)
125	FeaturesActivated	LONG	The set of flags that indicate which features have been purchased. (LCR600 only)
126	AvgFlowRate	VOLUME	Average flow rate.
127	CompFlowRate	VOLUME	Compensated flow rate.
128	AvgFlowBase	LIST+54	Specifies the time unit that is applied when calculating the flow rate.
129	AvgTemp	SFLOAT+1	The Average temperature during delivery.
130	AvgDensity	UFLOAT+4	The Average density of the fuel during delivery.
131	DensityUnits	TEXT	Specifies the unit that is used for density measurement.
132	MaxFlowRate	SFLOAT+1	Specifies the maximum flow rate used to calculate the corrected differential pressure.
133	MaxCorrectedPressure	SFLOAT+1	Specifies the corrected differential pressure relative to the current flow rate.
134		TEXT	Specifies the status of the density measurement.
Start of the LCR 600 POS Specific Fields			
192	PricePreset	SFLOAT+2	The price preset value for the current active product calibration number.
193	POSLockDiscounts	LIST+60	Specifies the current POS lock discounts setting.
194	POSLockPrice	LIST+61	Specifies the current POS lock price setting.
195	POSSaveNewPrice	LIST+62	Specifies the current POS save new price setting.
196	POSAllowNewDeliveryPrice	LIST+63	Specifies the current POS allow new delivery price setting.
197	POSProductNumber	INTEGER	Current POS product number.
198	POSProductName	TEXT	POS product name assigned to the current product number.
199	POSProductCode	TEXT	POS product code assigned to the current product number.
200	POSProductType	LIST+2	Type of POS product being delivered. Each product number can be assigned a unique product type.
201	POSProductCalibration	LIST+64	Calibration number assigned to the current POS product number.
202	POSProductPrice	SFLOAT+4	POS default price assigned to the current product number.
203	POSProductTax	LIST+65	POS tax category assigned to the current product number.
204	POSProductCashDiscount	LIST+66	POS cash discount assigned to the current product number.
205	POSProductVolumeDiscount	LIST+67	POS volume discount assigned to the current product number.
206	POSTaxNumber	LIST+68	Current POS tax number.
207	POSTaxName	TEXT	POS tax name assigned to the current tax number.

208	POSTaxCategory	LIST+69	POS tax category assigned to the current tax number.
209	POSTaxValue	SFLOAT+4	POS tax value assigned to the current tax number.
210	POSTaxType	LONG	POS per unit tax value assigned to the current tax number.
211	POSTaxHeader	TEXT	POS header tax value for the percent tax assigned to the current tax number.
212	POSCashDiscountNumber	LIST+71	Current POS cash discount number.
213	POSCashDiscountName	TEXT	POS cash discount name assigned to the current cash discount number.
214	POSCashDiscountInterval	LIST+72	POS cash discount interval assigned to the current cash discount number.
215	POSCashDiscountPercent	SFLOAT+3	POS cash discount percent assigned to the current cash discount number.
216	POSCashDiscountPerUnit	SFLOAT+4	POS per unit cash discount percent assigned to the current cash discount number.
217	POSCashDiscountDays	INTEGER	POS cash discount days interval assigned to the current cash discount number.
218	POSCashDiscountApply	LIST+73	POS cash discount apply field assigned to the current cash discount number.
219	POSVolumeDiscountNumber	LIST+74	Current POS volume discount number.
220	POSVolumeDiscountName	TEXT	POS volume discount name assigned to the current volume discount number.
221	POSVolumeDiscountInterval	LIST+75	POS volume discount interval assigned to the current volume discount number.
222	POSVolumeDiscountPercent	SFLOAT+3	POS volume discount percent assigned to the current volume discount number.
223	POSVolumeDiscountPerUnit	SFLOAT+4	POS per unit volume discount percent assigned to the current volume discount product number.
224	POSVolumeDiscountAmount	VOLUME	POS volume discount amount assigned to the current volume discount number.
225	POSVolumeDiscountApply	LIST+76	POS volume discount apply field assigned to the current volume discount number.
226	POSSubTotalDue	SFLOAT+2	Calculated subtotal on the current delivery.
227	POSPreVolumeDiscount	SFLOAT+2	Calculated volume discount available on the current delivery when applied before taxes.
228	POSTaxDue	SFLOAT+2	Calculated total tax due on the current delivery.
229	POSPostVolumeDiscount	SFLOAT+2	Calculated volume discount available on the current delivery applied after taxes.
230	POSTotalDue	SFLOAT+2	Calculated total due on the current delivery.
231	POSCashDiscount	SFLOAT+2	Calculated cash discount available on the current delivery.
232	POSDiscountDate	TEXT	Calculated discount date for a specific interval.
233	POSMiscChargeNumber	LIST+77	POS miscellaneous charge number assigned to the current miscellaneous charge number.
234	POSMiscChargeName	TEXT	POS miscellaneous charge name assigned to the current miscellaneous charge number.
235	POSMiscChargeTax	LIST+78	POS miscellaneous charge tax assigned to the current miscellaneous charge number.
236	POSMiscChargePrice	SFLOAT+4	POS miscellaneous charge price assigned to the current miscellaneous charge number.
237	POSMiscChargeQty	INTEGER	POS miscellaneous charge quantity assigned to the current miscellaneous charge number.

238	POSMiscChargeSubTotal	SFLOAT+2	Calculated POS miscellaneous charge subtotal on the current delivery.
239	POSMiscChargeTaxDue	SFLOAT+2	Calculated POS miscellaneous charge tax due on the on the current delivery.
240	POSMiscChargeTotal	SFLOAT+2	Calculated POS miscellaneous charge total due on the current delivery.
Start of LCR.iQ Specific Fields			
256	ActiveIOBoard	BYTE	The I/O board number being accessed by the LCR.iQ. The I/O board number is normalized to zero.
257	IOBoardUID	TEXT	UID associated with the active I/O board.
258	IOBoardName	TEXT	Name assigned to the active I/O board.
259	IOBoardNode	BYTE	Node address used by the LCR.iQ when communicating with the active I/O board.
260	PulserType	LIST1000+0	Specifies the type of pulser connected to the active I/O board.
261	MultipleSiteDelivery	LIST+5	Indicates that the next delivery will be made with the no-flow timer disabled. After the delivery, this field will be set back to LCRL_NO.
262	LastGrossQty	VOLUME	Contains the gross quantity printed on the last delivery ticket.
263	LastNetQty	VOLUME	Contains the net quantity printed on the last delivery ticket.
264	LastDeliveryQty	VOLUME	Contains the pertinent quantity printed on the last delivery ticket; it will contain the net quantity if the product delivered was temperature compensated, otherwise it will contain the gross quantity.
265	TimeFormat	LIST1000+1	Format to use when displaying the time.
266	MonetaryPrecision	LIST+14	Number of places past the decimal that are to be displayed in monetary fields.
267	MeterQmin	VOLUME	Minimum flow rate allowed through the meter. If the flow rate is greater than zero and falls below this value for the time specified in field #273, and the low flow rate is not due to the approaching end of a preset, the delivery will be paused with a warning to the fueler.
268	MeterQmax	VOLUME	Maximum flow rate allowed through the meter. If the flow rate exceeds this value for the time specified in field #274, the delivery will be paused with a warning to the fueler.
269	Jurisdiction	LIST1000+2	Specifies the W&M jurisdiction used when defining the defaults for protected calibration fields.
270	PresetRequired	LIST+5	Indicates that a preset is required for each delivery made.
271	PulserBehavior	LIST1000+4	Specifies how pulses are processed; they can be ignored, only forward pulses can be counted, only reverse pulses can be counted, or both forward and reverse pulses can be counted.
272	FuelingScreenColor	LIST1000+3	Defines the background color of the delivery active screen for the calibrated product.

273	LowFlowTimer	INTEGER	Used in conjunction with field #267 to determine when a delivery should be paused due to a low-flow condition.
274	HighFlowTimer	INTEGER	Used in conjunction with field #268 to determine when a delivery should be paused due to a high-flow condition.
275	MenuAccessProtection	LIST1000+5	Specifies the level of protection used when accessing the Main Menu or Setup Menu.
276	DeliveryTicket	LIST1000+6	Specifies which delivery ticket is to be used.
277	PrintMultipleDeliveries	LIST+5	Indicates whether or not the multiple deliveries at one site message is to be printed on the delivery ticket.
278	PrintAverageFlowRate	LIST+5	Indicates whether or not the average flow rate is to be printed on the delivery ticket.
279	PrintAverageTemperature	LIST+5	Indicates whether or not the average temperature is to be printed on the delivery ticket.
280	BlankLinesBeforeTicket	BYTE	Contains the number of blank lines that are to be printed before each delivery, shift, and calibration ticket.
281	BlankLinesAfterTicket	BYTE	Contains the number of blank lines that are to be printed after each delivery, shift, and calibration ticket.
282	HoseStuffingOption	LIST+5	Indicates whether or not the hose stuffing feature is to be used by the register.
283	IOBoardVersion	TEXT	Version of the SR1010 software running in the I/O board.
284	ValveLogicType	LIST1000+7	Valve logic used with presetting; either a two-stage preset valve, or a single stage / dual option preset valve.
285	PrintGrossQty	LIST+5	Indicates whether or not the gross volume is to be printed on the delivery ticket on net deliveries. Note that if Field #31 is set to 0, the settings of Field #285 and Field #286 are irrelevant; if Field #31 is set to 1, the two ticket text lines are controlled independently with Field #285 and Field #286.
286	PrintCompParam	LIST+5	Indicates whether or not the compensation parameter is to be printed on the delivery ticket on net deliveries. Note that if Field #31 is set to 0, the settings of Field #285 and Field #286 are irrelevant; if Field #31 is set to 1, the two ticket text lines are controlled independently with Field #285 and Field #286.

Field Data Types		
Type	Value	Description
TEXT	0	The text data type is an ASCIIZ text string.
INTEGER	1	The integer data type is a signed 2-byte integer.
DATE	2	The date data type is an ASCIIZ text string which contains the date using a MM/DD/YY or DD/MM/YY format, depending on the setting of Field #19.
TIME	3	The time data type is an ASCIIZ text string which contains the time using the "HH:MM:SS" format.
LONG	4	The long data type is a signed 4-byte integer.
VOLUME	5	The volume data type is a signed 4-byte integer. There is an implied decimal point associated with this data type which is defined by Field #39.
FFLOAT	6	The ffloat data type is a free format signed 4-byte floating point number in IEEE-754 format.
BYTE	7	The byte data type is an unsigned 1-byte integer.
LIST600+n	10	The list data type is an unsigned character which is an index into list table number n. See the following tables for the list types supported and for the indexes used for each list types.
LIST+n	50	The list data type is an unsigned character which is an index into list table number n. See the following tables for the list types supported and for the indexes used for each list types.
UFLOAT+n	80	The ufloat data type is a signed 4-byte floating point number which will always contain a positive number and displayed to n digits past the decimal point.
SFLOAT+n	90	The sfloat data type is a signed 4-byte floating point number and displayed to n digits past the decimal point.
LIST1000+n	100	The list data type is an unsigned character which is an index into list table number n. See the following tables for the list types supported and for the indexes used for each list types.

List Types	
List	Description
0	List group of calibration numbers: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or 16. Note: Calibration numbers 5 – 16 are only available when POS is enabled.
1	List group of presets allowed: None, Gross, Net, or Both.
2	List group of product types: Ammonia, Aviation, Distillate, Gasoline, Methanol, Lube Oil, or None.
3	List group of date formats: MM/DD/YY or DD/MM/YY.
4	List group of unit of measures: Gallons, Litres, Cubic Meters, Pounds, Kilograms, Barrels, or Other.
5	List group used for Yes and No selections: Yes or No.
6	List group of auxiliary unit of measures: Gallons, Litres, Cubic Meters, Pounds, Kilograms, Barrels, or Other.
7	List group preset types: Clear, Multiple, Retain, Inventory or Required.
8	List group of header line indexes: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12.
9	List group of pulse output edge options: Rising or Falling.
10	List group of temperature units: Degrees Celsius or Degrees Fahrenheit.
11	List group of rate base units: Per Second, Per Minute, or Per Hour.
12	List group of linearization point numbers: 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10.
13	List group of user selectable security level options: Locked or Unlocked.
14	List group of number of decimals used with volumes: Hundredths, Tenths, or Whole.
15	List group of compensation types: None, Linear F, Linear C, Table 24, Table 54, Table 6B, Table 54B, Table 54C, Table 54D, or NH3.
16	List group of diagnostic message indexes. These status conditions can also be retrieved via LCP02GetStatus.
17	List group of compensation parameters. Only used by ticket printing within the LCR.
18	List group of compensation temperatures. Only used by ticket printing within the LCR.
19	List group of printer types: Epson NewFontB, Epson NewFontA, Epson OldFontA, Epson OldFontB, Okidata Microline 184 Turbo or Blaster.
20	List group of residual processing options: Round or Truncate.
21	List group of flow direction options: Right or Left.
22	List group of auxiliary port 1 options: On During Delivery, Off, On, Monitor Flow Rate or Toggle Flow Rate.
23	List group of auxiliary port 2 options: Flow Direction, On During Delivery, Off, On, Monitor Flow Rate or Toggle Flow Rate.
24	List group of reset options: Clear All, No or Rebuild.
25	List group linearization activation state options: Applied or Setup.
26	List group of password usage options: None, Delivery or Shift.
27	List group of customer number usage options: None or Delivery.
47	List group of Print Gross/Parameter: Yes or No.
48	List group of Volume Corrected Message: Yes or No.
49	List group of Ticket Required: Yes, No or Skip.
50	List group of File Server Port: J1, J2 or J3.
51	List group of Baud Rate: 115200, 57600, 19200, 9600, 4800 or 2400.
52	List group of Tx Enable: None, !RTS, RTS, !DTR, DTR, !RTS!DTR or RTS!DTR.
53	List group of Delivery Screen: User Defined, Detailed Pump & Print, Volume & Preset, Aviation or Detailed POS.
60	List group of Lock Discount Category?: Yes or No.
61	List group of Lock Price?: Yes or No.
62	List group of \$ Change/Default?: Yes or No.
63	List group of Delivery \$ Change?: Yes or No.
64	List group of Calibration Number: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or 16.
65	List group of Tax Number: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or 16.
66	List group of Cash Discount: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or 16.

67	List group of Volume Number: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or 16.
68	List group of Tax Number: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or 16.
69	List group of Tax Category: A, B, C, D, E, F, G, H, I or J.
70	List group of Tax Type: Not Used, Percent, Per Unit or Tax on Tax.
71	List group of Cash Discount Number: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or 16.
72	List group of Cash Discount Interval: 1, 2 or 3.
73	List group of Apply Discount: Before Tax or After Tax.
74	List group of Volume Discount Number: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or 16.
75	List group of Volume Discount Interval: 1, 2 or 3.
76	List group of Apply Discount: Before Tax or After Tax.
77	List group of Miscellaneous Charge: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or 16.
78	List group of Miscellaneous Charge Tax: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, None or Same as Product.
100	List group of Pulser Types: No Pulser, All Single Channel, One Dual Channel, Two Dual Channel, One Triple Channel.
101	List group of Time Formats: 24 Hour, AM/PM.
102	List group of Jurisdictions: NTEP, Measurement Canada, OIML, EU.
103	List group of Fueling Screen Colors: Black, Blue, Green, Cyan, Red, Magenta, Brown, Light Gray, Gray, Light Blue, Light Green, Light Cyan, Light Red, Light Magenta, Yellow, White.
104	List group of Pulser Behaviors: Ignore, Count Forward, Count Reverse, Count All.
105	List group of Menu Access Protections: No Protection, Protect Main Menu, Protect Setup Menu.
106	List group of Delivery Tickets: No Ticket, Long Ticket, Short Ticket, Detailed Ticket, Custom Ticket.
107	List group of Valve Logic Types: Two Stage Valve, Single/Dual Option Valve.

List 0: Product Number

Value	Description
0	Product number 1 selected.
1	Product number 2 selected.
2	Product number 3 selected.
3	Product number 4 selected.
4	Product number 5 selected.
5	Product number 6 selected.
6	Product number 7 selected.
7	Product number 8 selected.
8	Product number 9 selected.
9	Product number 10 selected.
10	Product number 11 selected.
11	Product number 12 selected.
12	Product number 13 selected.
13	Product number 14 selected.
14	Product number 15 selected.
15	Product number 16 selected.

List 1: Preset Types Allowed	
Value	Description
0	No presets are allowed. The gross and net preset screens are not displayed on the lap pad and push button presets are not allowed.
1	Gross presets are allowed. The gross preset screen is displayed on the lap pad and push button presets are allowed if the product being delivered is not temperature compensated.
2	Net presets are allowed. The net preset screen is displayed on the lap pad and push button presets are allowed if the product being delivered is temperature compensated.
3	Both gross and net presets are allowed. The gross and net preset screens are displayed on the lap pad and push button presets are always allowed.

List 2: Product Type	
Value	Description
0	Product being delivered is ammonia.
1	Product being delivered is aviation fuel.
2	Product being delivered is a distillate, which is a family of hydrocarbon products.
3	Product being delivered is gasoline.
4	Product being delivered is methanol alcohol.
5	Product being delivered is liquefied petroleum gasoline, which is a mixture of propane and butane.
6	Product being delivered is lube oil, which is a family of heavy oils.
7	There is no product type associated with the product being delivered.

List 3: Date Format	
Value	Description
0	The date is displayed as MM/DD/YY.
1	The date is displayed as DD/MM/YY.

List 4: Units of Measure	
Value	Description
0	Product is delivered by the gallon.
1	Product is delivered by the litre.
2	Product is delivered by the cubic meter.
3	Product is delivered by the pound.
4	Product is delivered by the kilogram.
5	Product is delivered by the barrel.
6	The unit of measure for the product is one other than those listed above.

List 5: Yes/No Selection	
Value	Description
0	The functionality of the field has been selected.
1	The functionality of the field has been deselected.

List 6: Auxiliary Units of Measure

Value	Description
0	Auxiliary unit for product is delivered by the gallon.
1	Auxiliary unit for product is delivered by the litre.
2	Auxiliary unit for product is delivered by the cubic meter.
3	Auxiliary unit for product is delivered by the pound.
4	Auxiliary unit for product is delivered by the kilogram.
5	Auxiliary unit for product is delivered by the barrel.
6	The auxiliary unit of measure for the product is one other than those listed above.

List 7: Preset Types

Value	Description
0	Clear presets are those that will end the delivery when the preset amount is reached at which time the preset amount is cleared. This requires the user to explicitly enter the preset amount for each delivery.
1	Multiple presets will pause the delivery when the preset amount is reached and allow the same preset amount to be delivered again. The preset amount will be cleared when the delivery is explicitly ended with Command #0, or when it ends due to the no flow timer timing out.
2	Retain presets are similar to clear presets with the exception that the preset amount is not cleared at the end of the delivery.
3	Indicates that the preset fields are used as inventory fields. They are not reset between deliveries.
4	Required presets are similar to Clear presets in that they will end the delivery when the preset amount is reached at which time the preset amount is cleared. This requires the user to explicitly enter the preset amount for each delivery, and the preset amount cannot be set to 0 for this option.

List 8: Header Line Number

Value	Description
0	Header line 1 is accessible via Field #30.
1	Header line 2 is accessible via Field #30.
2	Header line 3 is accessible via Field #30.
3	Header line 4 is accessible via Field #30.
4	Header line 5 is accessible via Field #30.
5	Header line 6 is accessible via Field #30.
6	Header line 7 is accessible via Field #30.
7	Header line 8 is accessible via Field #30.
8	Header line 9 is accessible via Field #30.
9	Header line 10 is accessible via Field #30.
10	Header line 11 is accessible via Field #30.
11	Header line 12 is accessible via Field #30.

List 9: Pulse Output Edge

Value	Description
0	Indicates that a rising edge is generated on the scaled pulse output when the quantity is incremented.
1	Indicates that a falling edge is generated on the scaled pulse output when the quantity is incremented.

List 10: Temperature Units

Value	Description
0	Temperature calculations and displays are done in Celsius.
1	Temperature calculations and displays are done in Fahrenheit.

List 11: Rate Base

Value	Description
0	Rates are calculated and displayed in Field #38 per second.
1	Rates are calculated and displayed in Field #38 per minute.
2	Rates are calculated and displayed in Field #38 per hour.

List 12: Linearization Point Number

Value	Description
0	Data for linearization point 1 is accessible.
1	Data for linearization point 2 is accessible.
2	Data for linearization point 3 is accessible.
3	Data for linearization point 4 is accessible.
4	Data for linearization point 5 is accessible.
5	Data for linearization point 6 is accessible.
6	Data for linearization point 7 is accessible.
7	Data for linearization point 8 is accessible.
8	Data for linearization point 9 is accessible.
9	Data for linearization point 10 is accessible.

List 13: Security Status

Value	Description
0	The LCR unit is currently locked. If the switch on the LCR is not in the "six o'clock position," the unit will be at the delivery security level. If the switch is in the "six o'clock position," the unit will be at the weights and measures security level.
1	The LCR unit is currently unlocked. If the switch on the LCR is not in the "six o'clock position," the unit will be at the unlocked security level. If the switch is in the "six o'clock position," the unit will be at the weights and measures security level.

List 14: Number of Decimal Places

Value	Description
0	Fields with a type of VOLUME are calculated and displayed with two digits to the right of the decimal point.
1	Fields with a type of VOLUME are calculated and displayed with one digit to the right of the decimal point.
2	Fields with a type of VOLUME are calculated and displayed as an integer.

List 15: Compensation Types

Value	Description
0	The product being delivered is not temperature compensated.
1	Temperature compensation is being performed on the delivered product according to the linear F compensation rules.
2	Temperature compensation is being performed on the delivered product according to the linear C compensation rules.
3	Temperature compensation is being performed on the delivered product according to table 24 compensation rules.
4	Temperature compensation is being performed on the delivered product according to table 54 compensation rules.
5	Temperature compensation is being performed on the delivered product according to table 6B compensation rules.
6	Temperature compensation is being performed on the delivered product according to table 54B compensation rules.
7	Temperature compensation is being performed on the delivered product according to table 54C compensation rules.
8	Temperature compensation is being performed on the delivered product according to table 54D compensation rules.
9	Temperature compensation is being performed on the delivered product according to the NH3 compensation rules.

List 16: Diagnostic Messages

Value	Description
0	The checksum on the program ROM space has failed.
1	An error occurred while reading the temperature.
2	A watchdog timeout error occurred.
3	An invalid volume compensation type was detected.
4	An invalid volume compensation parameter was detected.
5	One of the volume compensation parameters was out of range.
6	A meter calibration error occurred.
7	The current delivery was ended due to a pulser failure.
8	The current delivery was ended due to the preset value being reached.
9	The current delivery was ended due to a no-flow timeout condition.
10	A request has been issued to stop (pause) the current delivery.
11	A request has been issued to end the current delivery.
12	A power fail condition was detected.
13	The current delivery is a preset delivery.
14	The terminal connected to the LCR when the delivery was started has been disconnected.
15	Field #37 is set to 0 and the printer attached to the RS-232 printer port is not ready.
16	An error occurred while accessing one of the data fields during a delivery.
17	A delivery ticket is pending. A new delivery cannot be started until this flag is cleared.
18	A shift ticket is pending.
19	The product being delivered is flowing through the meter.
20	A delivery is currently in progress.
21	A gross preset delivery is active.
22	A net preset delivery is active.
23	A gross preset delivery has stopped.
24	A net preset delivery has stopped.
25	Temperature volume compensation is active on the current delivery.
26	Solenoid 1 has been closed and the delivery is continuing in dwell mode.
29	A warning condition was detected during the initialization of a delivery.
30	End of list marker.

List 17: Compensation Parameters

Value	Description
0	The compensation parameter label printed on the delivery ticket is "NONE".
1	The compensation parameter label printed on the delivery ticket is "COEFFICIENT=".
2	The compensation parameter label printed on the delivery ticket is "COEFFICIENT=".
3	The compensation parameter label printed on the delivery ticket is "SG 60/60F=".
4	The compensation parameter label printed on the delivery ticket is "DENSITY KG/L=".
5	The compensation parameter label printed on the delivery ticket is "API GRAVITY=".
6	The compensation parameter label printed on the delivery ticket is "DENSITY KG/CuM=".
7	The compensation parameter label printed on the delivery ticket is "COEFFICIENT=".
8	The compensation parameter label printed on the delivery ticket is "DENSITY KG/CuM=".
9	The compensation parameter label printed on the delivery ticket is blank.

List 18: Compensation Temperatures

Value	Description
0	The label printed on the delivery ticket as the temperature unit is blank.
1	The label printed on the delivery ticket as the temperature unit is "°F".
2	The label printed on the delivery ticket as the temperature unit is "°C".
3	The label printed on the delivery ticket as the temperature unit is "°F".
4	The label printed on the delivery ticket as the temperature unit is "°C".
5	The label printed on the delivery ticket as the temperature unit is "°F".
6	The label printed on the delivery ticket as the temperature unit is "°C".
7	The label printed on the delivery ticket as the temperature unit is "°C".
8	The label printed on the delivery ticket as the temperature unit is "°C".
9	The label printed on the delivery ticket as the temperature unit is "°C".

List 19: Printer Type

Value	Description
0	An Epson NewFontB printer is connected to the RS-232 printer port.
1	An Epson NewFontA printer is connected to the RS-232 printer port.
2	An Epson OldFontA printer is connected to the RS-232 printer port.
3	An Epson OldFontB printer is connected to the RS-232 printer port.
4	An Okidata Microline 184 Turbo printer is connected to the RS-232 printer port.
5	A Blaster printer is connected to the RS-232 printer port.

List 20: Residual Processing

Value	Description
0	Volume residuals are rounded.
1	Volume residuals are truncated.

List 21: Flow Direction

Value	Description
0	The LCR is interpreting forward flow through the meter to be going to the right.
1	The LCR is interpreting forward flow through the meter to be going to the left.

List 22: Auxiliary Port 1 Options

Value	Description
0	The auxiliary port 1 will be turned on while a delivery is in progress. This also causes header line 11 to be printed on the delivery ticket, assuming the ticket is printed and the text for the header line is not blank.
1	Auxiliary port 1 is always off.
2	Auxiliary port 1 is always on.
3	Auxiliary port 1 is used to monitor the flow rate.
4	Auxiliary port 1 is used to toggle the flow rate.
5	Auxiliary port 1 is used to indicate pending transactions to be sent to the office.
6	Auxiliary port 1 is used to generate the pulse at the start of the delivery to signal the event to the external devices connected to the register,

List 23: Auxiliary Port 2 Options

Value	Description
0	The auxiliary port 2 signal is used in conjunction with the pulse output signal and indicates flow direction.
1	The auxiliary port 2 will be turned on while a delivery is in progress. This also causes header line 12 to be printed on the delivery ticket, assuming the ticket is printed and the text for the header line is not blank.
2	Auxiliary port 2 is always off.
3	Auxiliary port 2 is always on.
4	Auxiliary port 2 is used to toggle the flow rate.
5	Auxiliary port 2 is used to indicate that the density measurement is performed.
6	Auxiliary port 1 is used to indicate pending transactions to be sent to the office.
7	Auxiliary port 1 is used to generate the pulse at the start of the delivery to signal the event to the external devices connected to the register,

List 24: LCR Reset Options

Value	Description
0	All flash structures will be reset to their initial cleared values.
1	The functionality of the field has been deselected.
2	Each flash structure will be checked for containing valid data. The only structures cleared will be those containing errors.

List 25: Linearization Active State

Value	Description
0	The linearization points entered for the current product should be applied to deliveries of the product.
1	The linearization points entered for the current product should not be applied to deliveries of the product.

List 26: Password Usage (SR266)

Value	Description
0	The password field is not used.
1	The password field is required to start each new delivery.
2	The password field is required to start the first delivery in a new shift.

List 27: Customer Number Usage (SR266)

Value	Description
0	The customer number field is not used. It will be set to zero for all deliveries.
1	A prompt is issued for the customer number field before each new delivery.

List 49: Ticket Required

Value	Description
0	The ticket will be required to print after every delivery.
1	The ticket will not be required to print after every delivery. But if a printer is present, the ticket will print.
2	The ticket will not print after the delivery regardless of whether a printer is present or not.

List 50: File Server Port (SR60x)

Value	Description
0	J1 – Printer Port.
1	J2 – RS485 Port.
2	J3 – RS232 Port.

List 51: Baud Rate (SR60x)

Value	Description
0	Baud Rate 115200 for serial communication.
1	Baud Rate 57600 for serial communication.
2	Baud Rate 19200 for serial communication.
3	Baud Rate 9600 for serial communication.
4	Baud Rate 4800 for serial communication.
5	Baud Rate 2400 for serial communication.

List 52: Tx Enable (SR60x)

Value	Description
0	None TxEnable option for serial communication.
1	!RTS TxEnable option for serial communication.
2	RTS TxEnable option for serial communication.
3	!DTR TxEnable option for serial communication.
4	DTR TxEnable option for serial communication.
5	!RTS!DTR TxEnable option for serial communication.
6	RTSDTR TxEnable option for serial communication.

List 53: Delivery Screen (SR60x)

Value	Description
0	User defined – Not implemented.
1	Detailed Pump & Print – Detailed delivery totalizer and preset information is displayed during delivery.
2	Volume & Preset – Volume and Preset information is displayed during delivery.
3	Aviation – Aviation screen displayed during delivery.
4	Detailed POS – Detailed POS information is displayed during delivery.
5	Aviation & Weight – Aviation screen displaying weight and density of the fuel.
6	Delivery screen with the delivered quantity shown in the big digits during the active delivery.

List 60: Lock Discount Category (POS)

Value	Description
0	The POS discounts are locked and cannot be changed during a delivery.
1	The POS discounts are not locked and can be changed during a delivery.



List 61: Lock Price? (POS)	
Value	Description
0	The POS price amounts are locked and cannot be changed during a delivery.
1	The POS price amounts are not locked and can be changed during a delivery.

List 62: Price Change And Default? (POS)	
Value	Description
0	The POS price amounts can be changed for a delivery but will revert back to the default price amount for subsequent deliveries.
1	The POS price amounts can be changed for a delivery and will be set as the default price amount for subsequent deliveries.

List 63: Delivery Price Change? (POS)	
Value	Description
0	The POS price amounts can be changed during a delivery.
1	The POS price amounts cannot be changed during a delivery.

List 64: POS Product Numbers (POS)	
Value	Description
0	Calibration number 1 selected for the current POS product number.
1	Calibration number 2 selected for the current POS product number.
2	Calibration number 3 selected for the current POS product number.
3	Calibration number 4 selected for the current POS product number.
4	Calibration number 5 selected for the current POS product number.
5	Calibration number 6 selected for the current POS product number.
6	Calibration number 7 selected for the current POS product number.
7	Calibration number 8 selected for the current POS product number.
8	Calibration number 9 selected for the current POS product number.
9	Calibration number 10 selected for the current POS product number.
10	Calibration number 11 selected for the current POS product number.
11	Calibration number 12 selected for the current POS product number.
12	Calibration number 13 selected for the current POS product number.
13	Calibration number 14 selected for the current POS product number.
14	Calibration number 15 selected for the current POS product number.
15	Calibration number 16 selected for the current POS product number.

List 65: POS Tax Numbers (POS)	
Value	Description
0	POS Tax number 1 selected for the current POS product number.
1	POS Tax number 2 selected for the current POS product number.
2	POS Tax number 3 selected for the current POS product number.
3	POS Tax number 4 selected for the current POS product number.
4	POS Tax number 5 selected for the current POS product number.
5	POS Tax number 6 selected for the current POS product number.
6	POS Tax number 7 selected for the current POS product number.
7	POS Tax number 8 selected for the current POS product number.
8	POS Tax number 9 selected for the current POS product number.
9	POS Tax number 10 selected for the current POS product number.
10	POS Tax number 11 selected for the current POS product number.
11	POS Tax number 12 selected for the current POS product number.
12	POS Tax number 13 selected for the current POS product number.
13	POS Tax number 14 selected for the current POS product number.
14	POS Tax number 15 selected for the current POS product number.
15	POS Tax number 16 selected for the current POS product number.

List 66: POS Cash Discounts (POS)	
Value	Description
0	POS Cash Discount number 1 selected for the current POS product number.
1	POS Cash Discount number 2 selected for the current POS product number.
2	POS Cash Discount number 3 selected for the current POS product number.
3	POS Cash Discount number 4 selected for the current POS product number.
4	POS Cash Discount number 5 selected for the current POS product number.
5	POS Cash Discount number 6 selected for the current POS product number.
6	POS Cash Discount number 7 selected for the current POS product number.
7	POS Cash Discount number 8 selected for the current POS product number.
8	POS Cash Discount number 9 selected for the current POS product number.
9	POS Cash Discount number 10 selected for the current POS product number.
10	POS Cash Discount number 11 selected for the current POS product number.
11	POS Cash Discount number 12 selected for the current POS product number.
12	POS Cash Discount number 13 selected for the current POS product number.
13	POS Cash Discount number 14 selected for the current POS product number.
14	POS Cash Discount number 15 selected for the current POS product number.
15	POS Cash Discount number 16 selected for the current POS product number.

List 67: POS Volume Numbers (POS)	
Value	Description
0	POS Volume number 1 selected for the current POS product number.
1	POS Volume number 2 selected for the current POS product number.
2	POS Volume number 3 selected for the current POS product number.
3	POS Volume number 4 selected for the current POS product number.
4	POS Volume number 5 selected for the current POS product number.
5	POS Volume number 6 selected for the current POS product number.
6	POS Volume number 7 selected for the current POS product number.
7	POS Volume number 8 selected for the current POS product number.
8	POS Volume number 9 selected for the current POS product number.
9	POS Volume number 10 selected for the current POS product number.
10	POS Volume number 11 selected for the current POS product number.
11	POS Volume number 12 selected for the current POS product number.
12	POS Volume number 13 selected for the current POS product number.
13	POS Volume number 14 selected for the current POS product number.
14	POS Volume number 15 selected for the current POS product number.
15	POS Volume number 16 selected for the current POS product number.

List 68: POS Tax Numbers (POS)	
Value	Description
0	POS Tax number 1 selected.
1	POS Tax number 2 selected.
2	POS Tax number 3 selected.
3	POS Tax number 4 selected.
4	POS Tax number 5 selected.
5	POS Tax number 6 selected.
6	POS Tax number 7 selected.
7	POS Tax number 8 selected.
8	POS Tax number 9 selected.
9	POS Tax number 10 selected.
10	POS Tax number 11 selected.
11	POS Tax number 12 selected.
12	POS Tax number 13 selected.
13	POS Tax number 14 selected.
14	POS Tax number 15 selected.
15	POS Tax number 16 selected.

List 69: POS Tax Categories (POS)	
Value	Description
0	POS Category A selected.
1	POS Category B selected.
2	POS Category C selected.
3	POS Category D selected.
4	POS Category E selected.
5	POS Category F selected.
6	POS Category G selected.
7	POS Category H selected.
8	POS Category I selected.
9	POS Category J selected.

List 70: POS Tax Types (POS)	
Value	Description
0	POS Tax Type is not used.
1	POS Tax Type is used as a percentage value.
2	POS Tax Type is used as a per unit value.
3	POS Tax Type is used as a tax on tax value and can be based off of other tax category definitions.

List 71: POS Cash Discount Numbers (POS)	
Value	Description
0	POS Cash Discount number 1 selected.
1	POS Cash Discount number 2 selected.
2	POS Cash Discount number 3 selected.
3	POS Cash Discount number 4 selected.
4	POS Cash Discount number 5 selected.
5	POS Cash Discount number 6 selected.
6	POS Cash Discount number 7 selected.
7	POS Cash Discount number 8 selected.
8	POS Cash Discount number 9 selected.
9	POS Cash Discount number 10 selected.
10	POS Cash Discount number 11 selected.
11	POS Cash Discount number 12 selected.
12	POS Cash Discount number 13 selected.
13	POS Cash Discount number 14 selected.
14	POS Cash Discount number 15 selected.
15	POS Cash Discount number 16 selected.

List 72: POS Cash Discount Interval (POS)	
Value	Description
0	POS Cash Discount interval 1 selected.
1	POS Cash Discount interval 2 selected.
2	POS Cash Discount interval 3 selected.

List 73: POS Apply Cash Discount (POS)	
Value	Description
0	POS Cash Discount is applied before tax.
1	POS Cash Discount is applied after tax.

List 74: POS Volume Discount Numbers (POS)	
Value	Description
0	POS Volume Discount number 1 selected.
1	POS Volume Discount number 2 selected.
2	POS Volume Discount number 3 selected.
3	POS Volume Discount number 4 selected.
4	POS Volume Discount number 5 selected.
5	POS Volume Discount number 6 selected.
6	POS Volume Discount number 7 selected.
7	POS Volume Discount number 8 selected.
8	POS Volume Discount number 9 selected.
9	POS Volume Discount number 10 selected.
10	POS Volume Discount number 11 selected.
11	POS Volume Discount number 12 selected.
12	POS Volume Discount number 13 selected.
13	POS Volume Discount number 14 selected.
14	POS Volume Discount number 15 selected.
15	POS Volume Discount number 16 selected.

List 75: POS Volume Discount Interval (POS)	
Value	Description
0	POS Volume Discount interval 1 selected.
1	POS Volume Discount interval 2 selected.
2	POS Volume Discount interval 3 selected.

List 76: POS Apply Volume Discount (POS)	
Value	Description
0	POS Volume Discount is applied before tax.
1	POS Volume Discount is applied after tax.

List 77: Miscellaneous Charge Numbers (POS)	
Value	Description
0	POS Miscellaneous Charge number 1 selected.
1	POS Miscellaneous Charge number 2 selected.
2	POS Miscellaneous Charge number 3 selected.
3	POS Miscellaneous Charge number 4 selected.
4	POS Miscellaneous Charge number 5 selected.
5	POS Miscellaneous Charge number 6 selected.
6	POS Miscellaneous Charge number 7 selected.
7	POS Miscellaneous Charge number 8 selected.
8	POS Miscellaneous Charge number 9 selected.
9	POS Miscellaneous Charge number 10 selected.
10	POS Miscellaneous Charge number 11 selected.
11	POS Miscellaneous Charge number 12 selected.
12	POS Miscellaneous Charge number 13 selected.
13	POS Miscellaneous Charge number 14 selected.
14	POS Miscellaneous Charge number 15 selected.
15	POS Miscellaneous Charge number 16 selected.

List 78: Miscellaneous Charge Tax (POS)	
Value	Description
0	POS Miscellaneous Charge tax 1 selected.
1	POS Miscellaneous Charge tax 2 selected.
2	POS Miscellaneous Charge tax 3 selected.
3	POS Miscellaneous Charge tax 4 selected.
4	POS Miscellaneous Charge tax 5 selected.
5	POS Miscellaneous Charge tax 6 selected.
6	POS Miscellaneous Charge tax 7 selected.
7	POS Miscellaneous Charge tax 8 selected.
8	POS Miscellaneous Charge tax 9 selected.
9	POS Miscellaneous Charge tax 10 selected.
10	POS Miscellaneous Charge tax 11 selected.
11	POS Miscellaneous Charge tax 12 selected.
12	POS Miscellaneous Charge tax 13 selected.
13	POS Miscellaneous Charge tax 14 selected.
14	POS Miscellaneous Charge tax 15 selected.
15	POS Miscellaneous Charge tax 16 selected.
16	POS Miscellaneous Charge tax is set to None.
17	POS Miscellaneous Charge tax is set to the same value as the Product.

List 100: Pulser Type (LCR.iQ)	
Value	Description
0	No pulser is connected to the register.
1	All pulsers connected to the register are single channel.
2	One dual channel pulser is connected to the register.
3	Two dual channel pulsers are connected to the register.
4	One triple channel pulser is connected to the register.

List 101: Time Format (LCR.iQ)	
Value	Description
0	The time is displayed as a 24-hour clock, i.e. HH:MM:SS.
1	The time is displayed in AM/PM format, i.e. HH:MM AM or HH:MM PM.

List 102: Jurisdiction (LCR.iQ)	
Value	Description
0	NTEP.
1	Measurement Canada.
2	OIML.
3	EU.

List 103: Fueling Screen Color (LCR.iQ)	
Value	Description
0	Black is used on the fueling screen.
1	Blue is used on the fueling screen.
2	Green is used on the fueling screen.
3	Cyan is used on the fueling screen.
4	Red is used on the fueling screen.
5	Magenta is used on the fueling screen.
6	Brown is used on the fueling screen.
7	Light gray is used on the fueling screen.
8	Gray is used on the fueling screen.
9	Light blue is used on the fueling screen.
10	Light green is used on the fueling screen.
11	Light cyan is used on the fueling screen.
12	Light red is used on the fueling screen.
13	Light magenta is used on the fueling screen.
14	Yellow is used on the fueling screen.
15	White is used on the fueling screen.

List 104: Pulser Behavior (LCR.iQ)	
Value	Description
0	The pulser is ignored.
1	Forward pulses are counted.
2	Reverse pulses are counted.
3	All pulses are counted.

List 105: Menu Access Protection (LCR.iQ)	
Value	Description
0	No protection is provided for menu access.
1	Protection is provided when attempting to access the Main Menu.
2	Protection is provided when attempting to access the Setup Menu.

List 106: Delivery Ticket (LCR.iQ)	
Value	Description
0	No ticket is printed.
1	The Long ticket is printed.
2	The Short ticket is printed.
3	The Detailed ticket is printed.
4	The Custom ticket is printed.

List 107: Valve Logic Type (LCR.iQ)	
Value	Description
0	The value logic is a two-stage preset valve.
1	The valve logic is a single/dual option preset valve.

Delivery Code Bits	
Value	Description
0x0001	Indicates that a delivery ticket is pending. A new delivery cannot be started until this bit is cleared via successfully printing the last delivery ticket.
0x0002	Indicates that a shift ticket has been requested and is waiting to be printed.
0x0004	Indicates flow is active during a delivery. The bit is turned on and off with the Delivery Code Bit 0x0008 but is also turned off when a delivery is paused and turned back on when it is resumed.
0x0008	Indicates that a delivery is active. The bit is turned on just before Delivery Code Bit #10 is turned off and is turned off at the end of the delivery.
0x0010	Indicates the current delivery is delivering a gross preset quantity. When this quantity is reached the delivery will either end or pause depending on the value of Field #27.
0x0020	Indicates the current delivery is delivering a net preset quantity. When this quantity is reached the delivery will either end or pause depending on the value of Field #27.
0x0040	Indicates the current delivery has been stopped due to the gross preset value being reached. Depending on the value of Field #27 the delivery will either terminate or pause.
0x0080	Indicates the current delivery has been stopped due to the net preset value being reached. Depending on the value of Field #27 the delivery will either terminate or pause.
0x0100	Indicates that the current product will be temperature volume compensated during delivery.
0x0200	This bit is cleared at the beginning of a delivery and is set when solenoid 1 is closed due to the remaining gross or net preset value being less than or equal to the value in Field #26.
0x0400	Indicates that a delivery is in the process of being started. Once the delivery has been started successfully, this bit is turned off.
0x0800	Indicates that a new delivery has been queued in the LCR. This condition occurs when a Command #0 is issued and the switch is in the STOP, PRINT, or SHIFT PRINT position.
0x1000	Indicates that a data access error occurred during a delivery which was not critical to the delivery but that a default was used in place of the data that was attempting to be read.
0x2000	Indicates that a configuration event has occurred.
0x4000	Indicates that a calibration event has occurred.
0x8000	Indicates that the transaction record has been saved into the NV memory.

Delivery Status Bits	
Value	Description
0x0001	Indicates the delivery could not be started due to the checksum of the program code space failing.
0x0002	Indicates that the current delivery was not started or was terminated due to an error detected with the temperature reading hardware. The current delivery will either not be started or will terminate if this bit is set.
0x0004	Indicates that the processor was reset due to a watchdog timeout error. If a delivery was active, it was terminated.
0x0008	Indicates that there was an error setting up the volume compensation factor for the current product.
0x0010	Indicates the temperature of the product being delivered is outside the valid range for the compensation type of the current product.
0x0020	Indicates an error occurred while setting up the meter calibration for the current delivery. A delivery cannot be started if this bit is set.
0x0040	Indicates the current delivery was terminated due to too many pulser reversals.
0x0080	Indicates that a gross or net preset value has been reached. Depending on the value of Field #27 the delivery will either terminate or pause.
0x0100	Indicates the current delivery was stopped due to no flow being detected through the meter for the amount of time specified by Field #25.
0x0200	This bit is set if a delivery is active and Command #1 is issued. It is cleared when a delivery is started or resumed or when Command #2 or Command #6 has been issued.
0x0400	Indicates a request has been issued to end the current delivery. This occurs when a delivery is active and Command #2 or Command #6 has been issued.
0x0800	Indicates the current delivery was terminated due to a power failure for more than 15 seconds.
0x1000	Indicates that an error occurred while accessing one of the fields associated with preset deliveries. A delivery cannot be started or continue if this bit is set.
0x2000	Indicates the RS-232 lap pad has been disconnected from the LCR during a delivery. This bit is not used in RS-485 mode.
0x4000	Indicates that the current delivery requires a ticket to be printed and that the printer reports it is off line or busy. A delivery cannot be started if this bit is set.
0x8000	Indicates that a data access error occurred during a delivery which was critical to the delivery. The delivery was either not started or was terminated due to the error.

Machine Code Bits	
Value	Description
0x?0	The switch is in between positions.
0x01	The switch is in the run position.
0x02	The switch is in the stop position.
0x03	The switch is in the print position.
0x04	The switch is in the shift-print position.
0x05	The switch is in the calibrate position.
0x07	Mask that can be used to isolate the switch position.
0x07	The actual status of the LCR is not available. This status is returned if a broadcast message is issued.
0x08	The RS-232 printer is printing.
0x0?	The run state is active when a delivery has been started and flow is active.
0x10	The stop state is active when a delivery has been started but no flow is active.
0x20	The end delivery state is active when no delivery is active and no flow is active.
0x30	The auxiliary state is active when the switch has been in the shift-print position for less than two seconds since the last delivery was completed or if Command #3 is issued. If a delivery is not active, a calibration/diagnostic ticket can be printed by issuing Command #6.
0x40	The shift state is active when the switch has been in the shift-print position for at least two seconds since the last delivery was completed or if Command #4 is issued. If a delivery is not active, a shift ticket can be printed by issuing Command #6.
0x50	The calibrate state is active when no delivery is active, no flow is active, and the switch is in the calibrate position. If the LCR was previously in factory mode this state can be activated by issuing Command #5.
0x60	The wait for no flow state is active when a delivery is being stopped but flow is still active.
0x70	Mask that can be used to isolate the machine state.
0x80	Flag indicating an error has occurred in the LCR. The delivery, printer, and hardware status fields should be checked for details.

Printer Bits	
Value	Description
0x01	This bit is set when a delivery ticket print request is issued and is cleared when Printer Bit #7 is set indicating the request is being processed.
0x02	This bit is set when a shift ticket print request is issued and is cleared when Printer Bit #7 is set indicating the request is being processed.
0x04	This bit is set when a diagnostic ticket print request is issued and is cleared when Printer Bit #7 is set indicating the request is being processed.
0x08	This bit is set when a user print request is issued and is cleared when Printer Bit #7 is set indicating the request is being processed.
0x10	This bit is set when if the printer is out of paper.
0x20	This bit is set when there is no print processor online.
0x40	This bit is set when an error was detected with the print processor itself or if a data access error occurred while processing the current print request.
0x80	Indicates the print processor has begun to print a ticket or user print request.

Security Levels	
Value	Description
0x00	Indicates that a delivery is active but has been paused.
0x01	Indicates that the switch is out of calibration, the LCR is not unlocked, and no delivery is active.
0x02	Indicates that the switch is out of calibration, the LCR is unlocked, and no delivery is active.
0x03	Indicates that the switch is in calibration, the factory key has not been entered, and a delivery is not active.
0x04	Indicates that the switch is in calibration, the factory key has been entered correctly, and no delivery is active.
0x08	The field is always editable. This is never returned by the "Get Security Level" message (MsgID 27h). It is only returned as an edit security level by the "Get Field Parameters" message (MsgID 7Fh).
0x09	The field is never editable. This is never returned by the "Get Security Level" message (MsgID 27h). It is only returned as an edit security level by the "Get Field Parameters" message (MsgID 7Fh).
0x80	This bit is set if a delivery is active. Note that this makes the security level negative thus indicating that no fields are editable with the exception of those that are set to 0x08.

LCR A to D Code Masks	
Value	Description
0x01	Indicates that temperature calibration has been performed with a 100.0 0.01% ohm resistor.
0x02	Indicates that temperature calibration has been performed with a 128.6 0.01% ohm resistor.
0x04	Indicates that the temperature slope was entered explicitly.
0x08	Indicates that the temperature offset was entered explicitly.
0x10	Indicates that voltage calibration has been performed with 12 volts applied to the LCR.
0x20	Indicates that voltage calibration has been performed with 16 volts applied to the LCR.
0x40	Indicates that the voltage slope was entered explicitly.
0x80	Indicates that the voltage offset was entered explicitly.

Command Codes	
Command	Description
0	Start/Resume a delivery.
1	Pause the current delivery.
2	End the current delivery and print the ticket (if settings allow).
3	Place the LCR in auxiliary state.
4	Place the LCR in shift state and print the ticket (if settings allow).
5	Place the LCR in calibration state. This only applies if the LCR was previously in factory mode.
6	Print a ticket based on the current state of the LCR. This command does not end any deliveries, nor will it print a ticket if a delivery is active.

Return Codes	
Value	Description
0	The LCR request was completed successfully.
32	An invalid parameter ID was received by the LCR device. This can only occur during a LCP02Open operation when the field width and type tables are being initialized.
33	An invalid field number was detected by the LCR device.
34	The LCR device detected that the field data received is not compatible with the field data type.
35	The requested field was not set due to the mode of the LCR.
36	An invalid command number was detected by the LCR device.
37	An invalid device address was detected by the LCR device.
38	The last request made to a LCR has been queued by the LCR.
39	The LCR has determined there is no queued request in the LCR.
40	The queued request in the LCR was successfully aborted.
41	The previous request to the LCR, which was queued and then aborted, is still being processed by the LCR.
42	The queued request in the LCR has reached a point in its processing where it can no longer be aborted.
43	An invalid parameter block number was requested by LCP02Open.
44	An invalid baud rate was requested by LCP02Open.
48	The START operation failed while accessing the flash memory.
49	No acknowledge was received while accessing flash memory.
50	A CRC error was detected by the flash memory access routines.
51	The flash memory access routines received invalid parameters from the calling application.
52	An attempt to write over a page boundary was detected by the flash memory access routines.
53	The data written to flash memory failed the read-back CRC test.
64	The A to D converter is returning a value that is greater than its maximum valid value.
65	The A to D converter is returning a value that is less than its maximum valid value.
67	The compensation type, Field #57, contains an invalid value.
68	The compensation parameter in Field #58 is out of its valid limit range.
69	The current temperature of the product being delivered is outside the allowed compensation range.
80	A NBS card setup error occurred.
81	A NBS card termination error occurred.
82	Connection with the specified device failed.
83	The message received by the C-API from the specified device was the incorrect size.
84	An invalid device number was detected by the C-API.
85	An invalid field number was detected by the C-API.
86	The C-API detected that the field data received is not compatible with the field data type.
87	An invalid command number was detected by the C-API.
88	There are no devices attached to the LCR network.
89	The major version number of the C-API does not match the major version number of all LCR devices in the network.
90	The DOS environment parameter string does not exist or contains invalid data.
91	The C-API has determined there is no queued request in a LCR.
92	The C-API has determined that a request is already queued in a LCR.
93	An invalid request code has been queued by the C-API. This typically means the PC-based application has corrupted its data segment.
94	Network communications has not been opened with a call to LCP02Open.
95	The network communications has already been opened.
96	The printer is busy.
97	The text string to be printed overflowed the print buffer.
98	The requested status from the printer was not returned from the printer within the allotted time.

112	The requested field has not been initialized in flash memory.
113	The new field data failed its range check.
114	The field data cannot be read from flash memory.
115	A gross preset delivery is not active.
116	A net preset delivery is not active.
117	Set operations are never allowed on the requested field.
118	The requested field is no longer used.
119	The current switch position does not allow the requested operation to be performed.
120	The current state of the LCR does not allow the requested operation to be performed.
121	The requested command was ignored due to a pending delivery ticket.
122	A write to flash memory was not allowed due to hardware protection via the switch on the LCR.
123	The entered flow rate for the current linearization point is already being used by another point. Duplicate flow rates are not allowed.
124	After the linearization points were sorted in descending order it was detected that the percent error of two adjacent points differed by more than 0.25%. Adjacent linearization points cannot differ by more than 0.25%.
125	The end of the transaction list has been reached.
126	The end of the transaction list has been reached and some transactions were lost due to the queue filling up before the transactions were retrieved.
127	The customer number has not been set correctly.
200	The low level LCP drivers are already installed.
201	An invalid name and revision string was received by the LCP drivers.
202	An invalid node address was received by the LCP drivers.
203	An invalid base communications address was received by the LCP drivers.
204	An invalid IRQ was received by the LCP drivers.
205	An invalid baud rate was received by the LCP drivers.
206	An invalid timeout value was received by the LCP drivers.
207	An invalid transmit enable bit was received by the LCP drivers.
208	A NULL pointer was passed as an address to the LCP drivers.
209	The LCR reported that no request was active.
210	The LCR did not respond within the timeout value specified.
211	The LCR aborted the last request.
212	The LCR reported that the request made is not supported.
213	The LCR reported a buffer overflow condition.
214	The LCR reported that it is busy working on the last request.
215	The LCP drivers are not installed.
216	The LCP drivers detected a dynamic memory allocation error.
217	The clock speed of the CPU was not set for the LCP drivers (obsolete).
218	The message size of the incoming message does not match the expected size.
219	There was a problem initializing WinSock.
220	Error initializing socket.
221	An invalid IP address has been passed in.
222	An invalid IP port number has been passed in.
223	The message with an invalid message ID has been received.